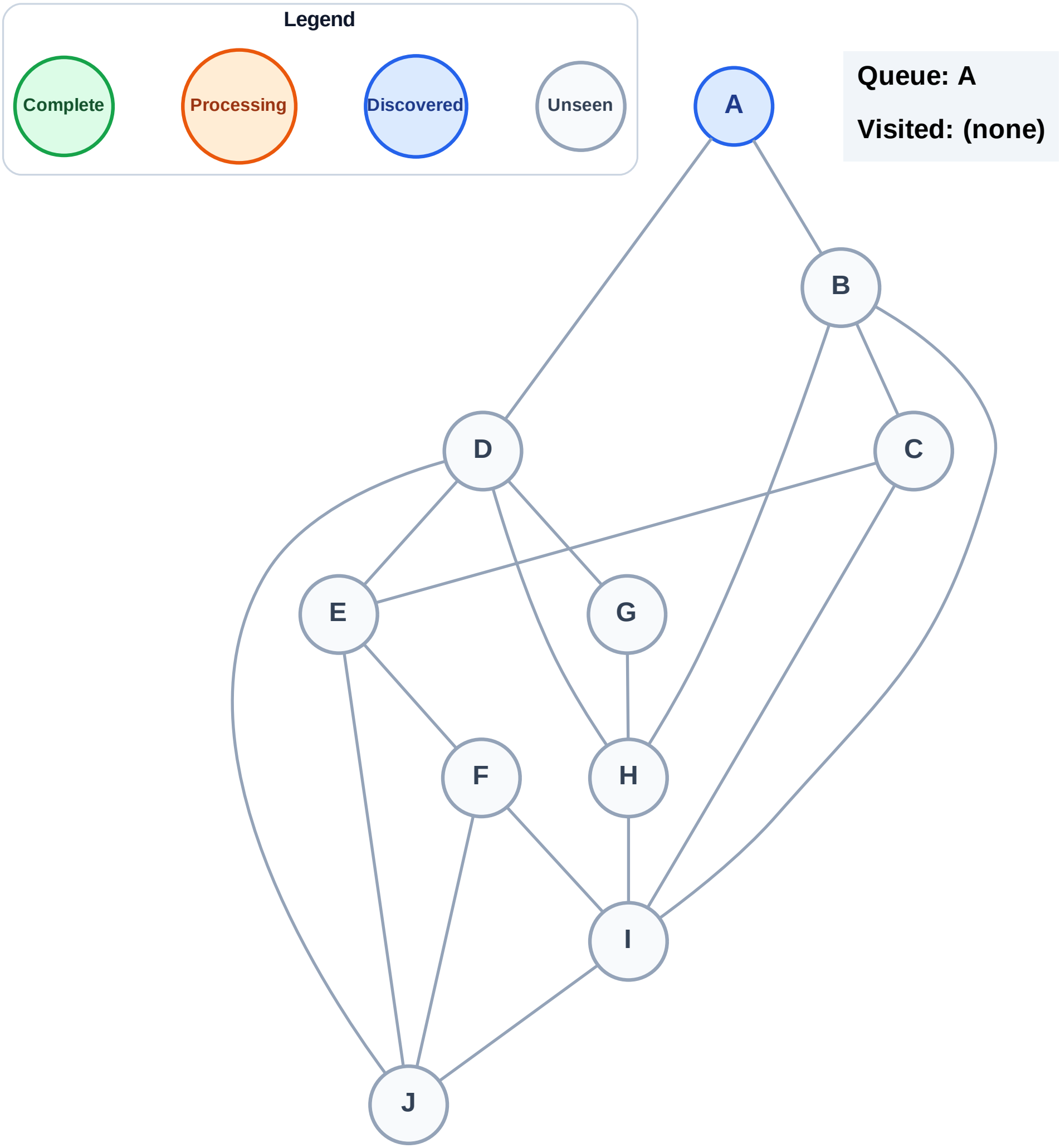
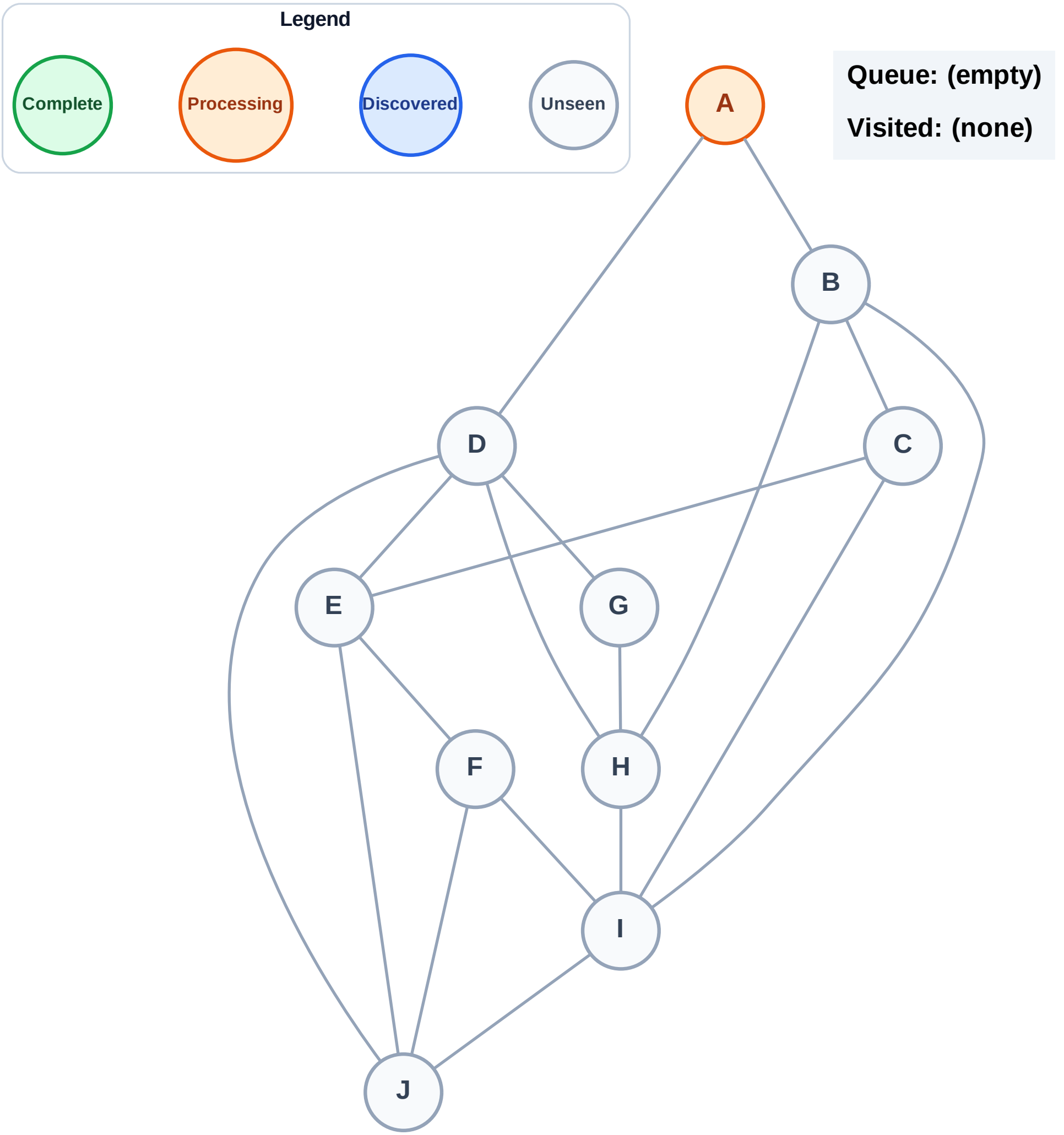
BFS Traversal

Step 1: Start at A



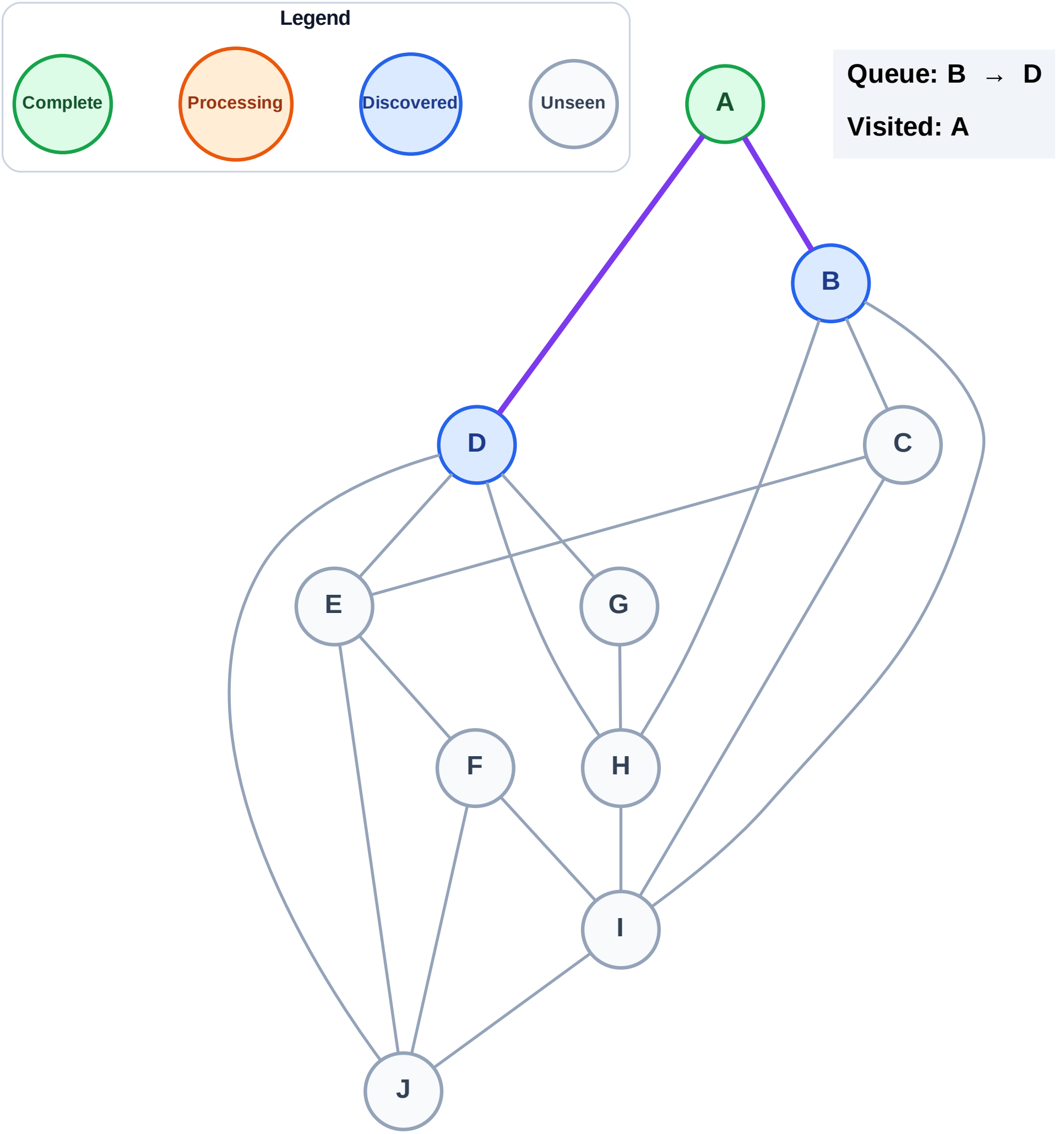
BFS Traversal

Step 2: Process node A



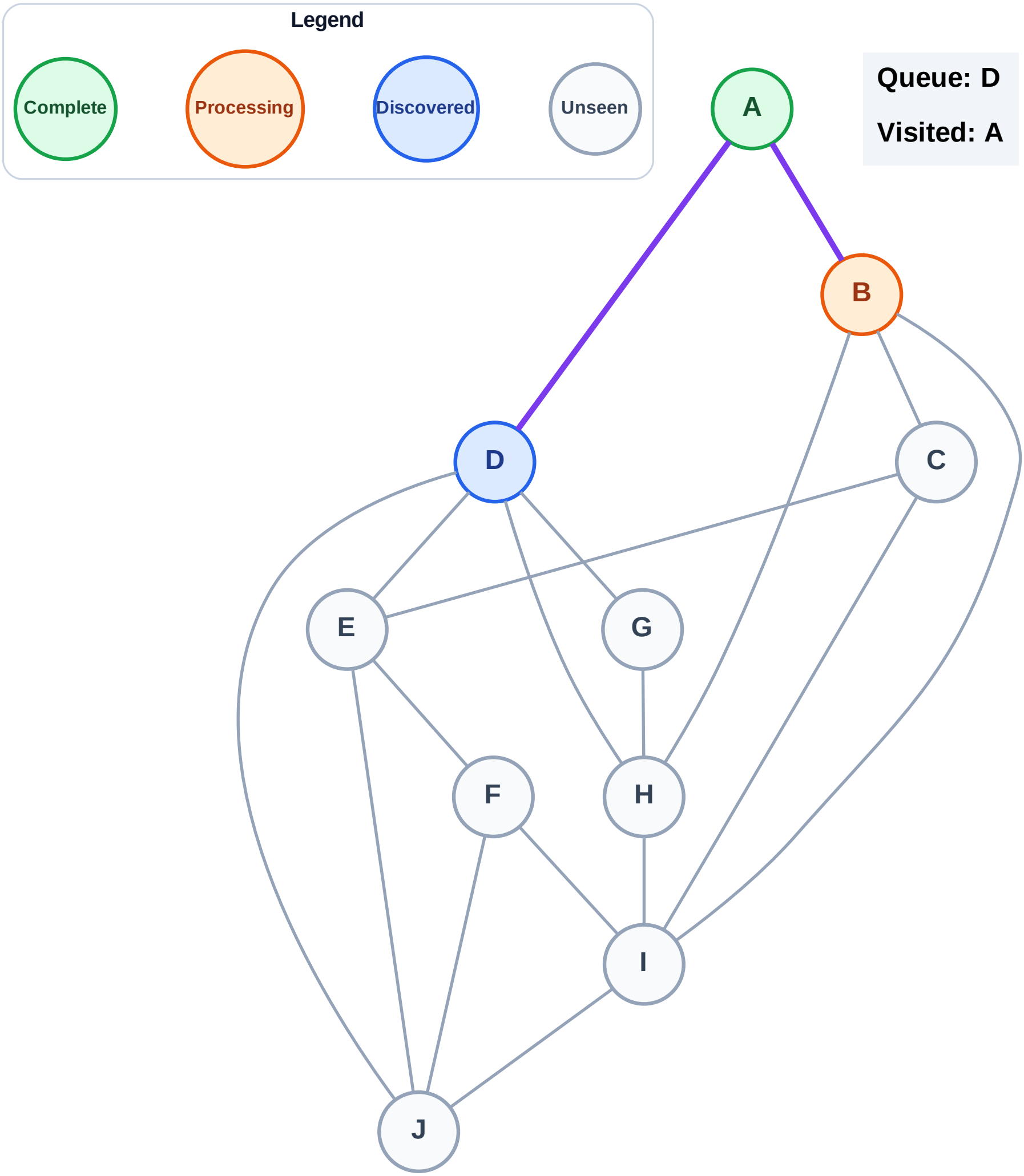
BFS Traversal

Step 3: Discover B, D from A



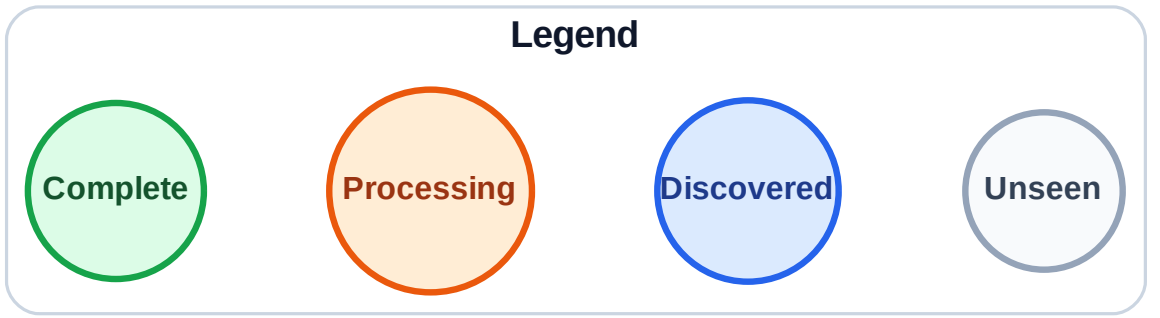
BFS Traversal

Step 4: Process node B



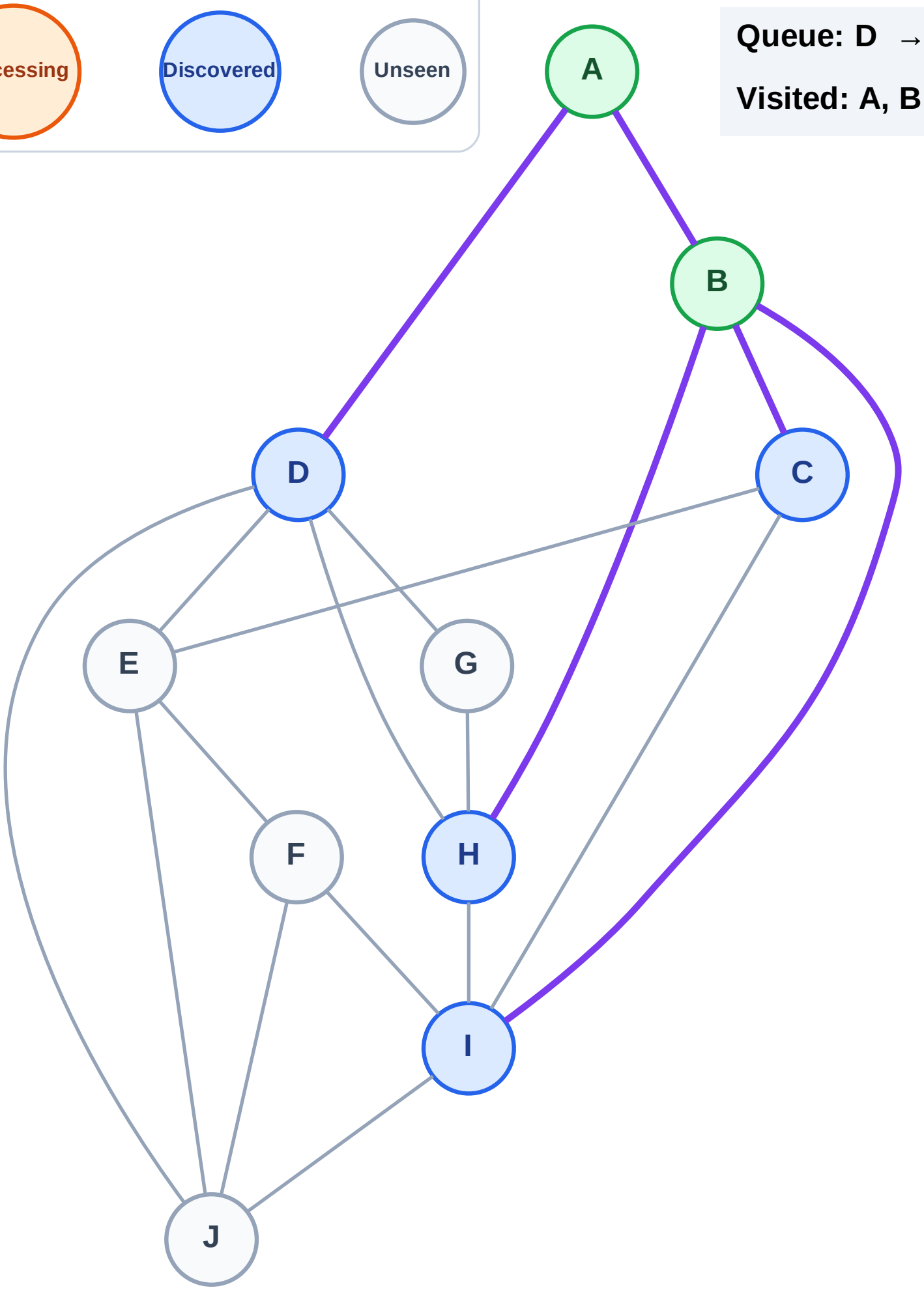
BFS Traversal

Step 5: Discover C, H, I from B



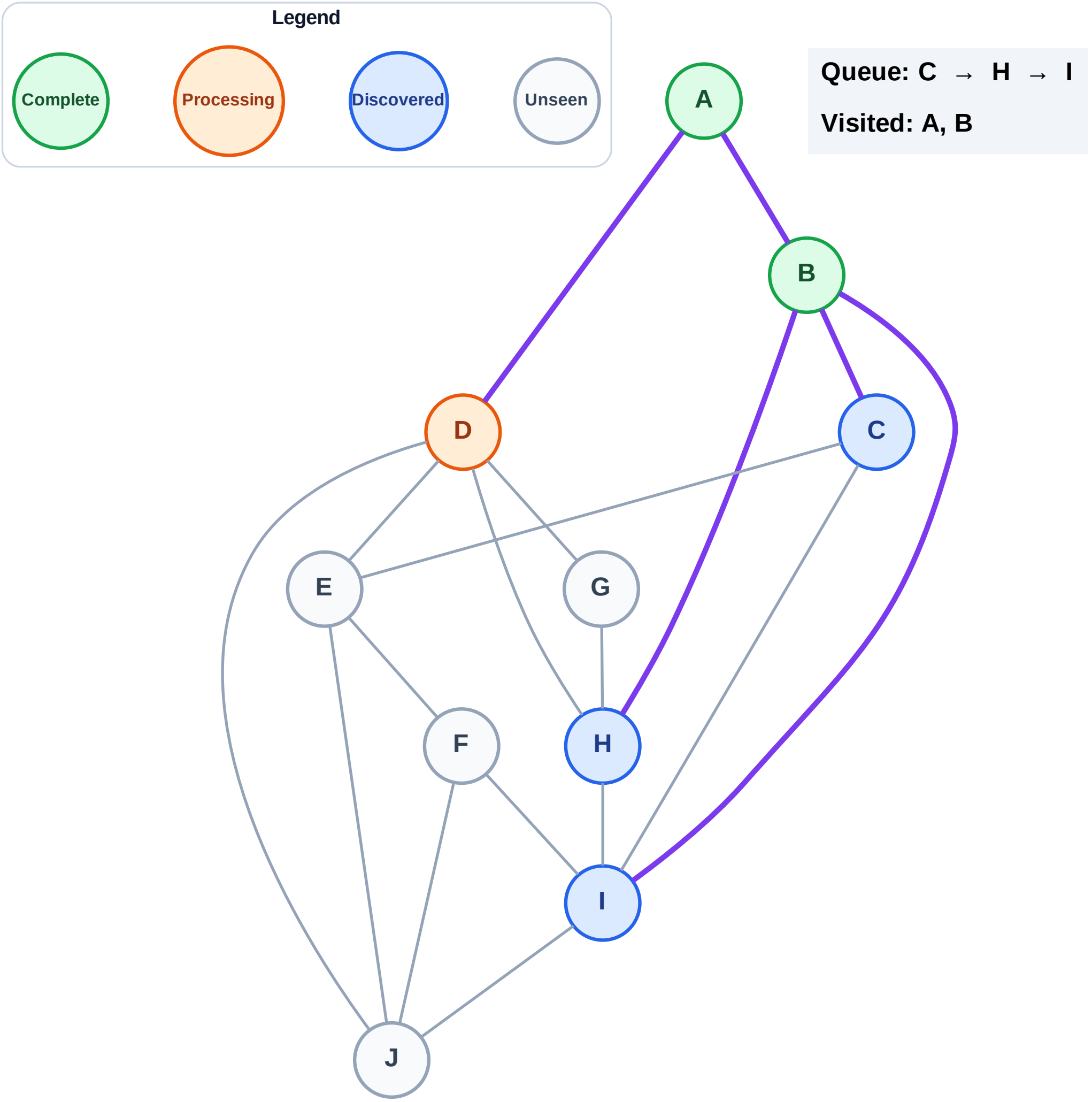
Queue: D → C → H → I

Visited: A, B



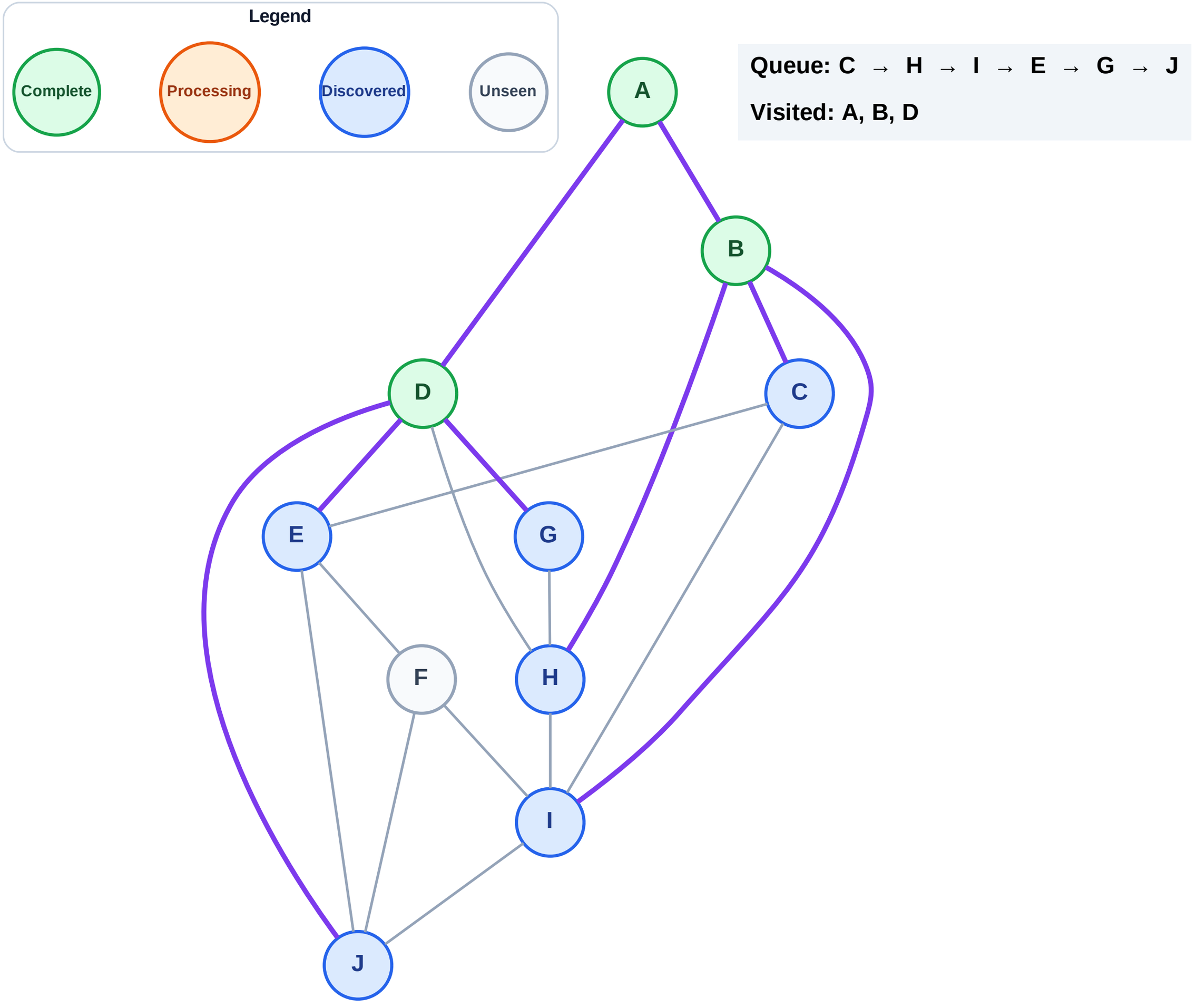
BFS Traversal

Step 6: Process node D



BFS Traversal

Step 7: Discover E, G, J from D

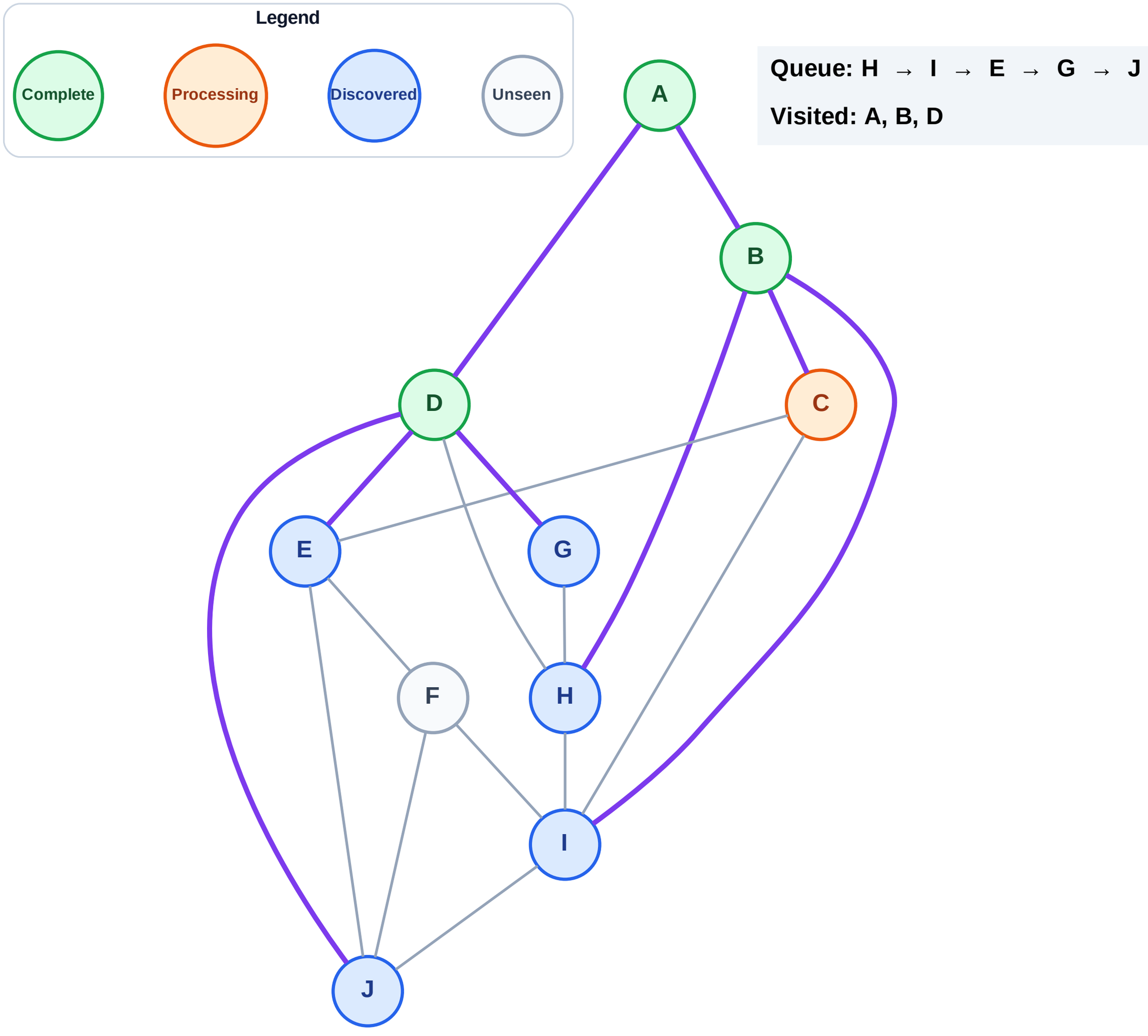


Queue: C → H → I → E → G → J

Visited: A, B, D

BFS Traversal

Step 8: Process node C

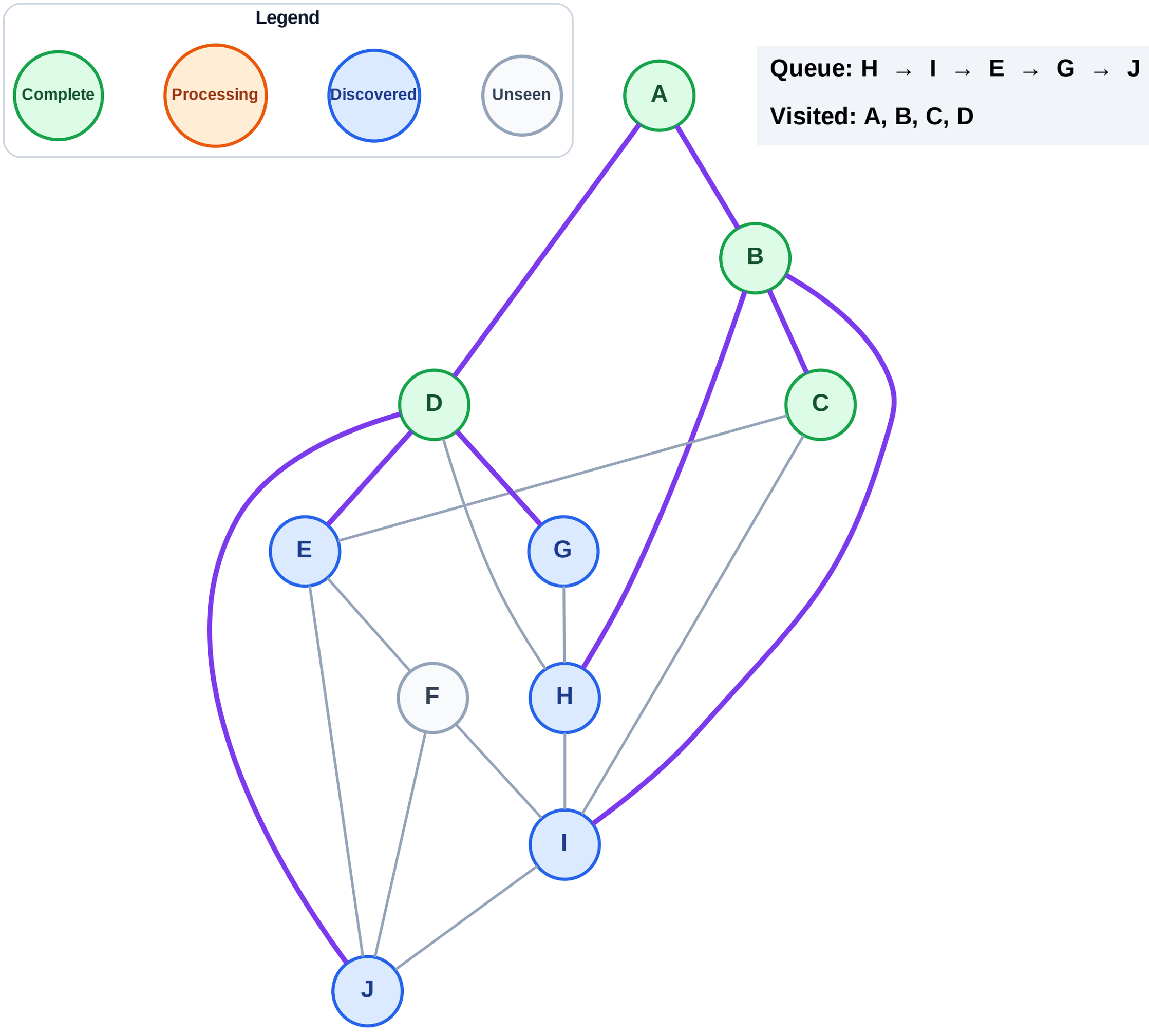


Queue: H → I → E → G → J

Visited: A, B, D

BFS Traversal

Step 9: No new neighbors from C

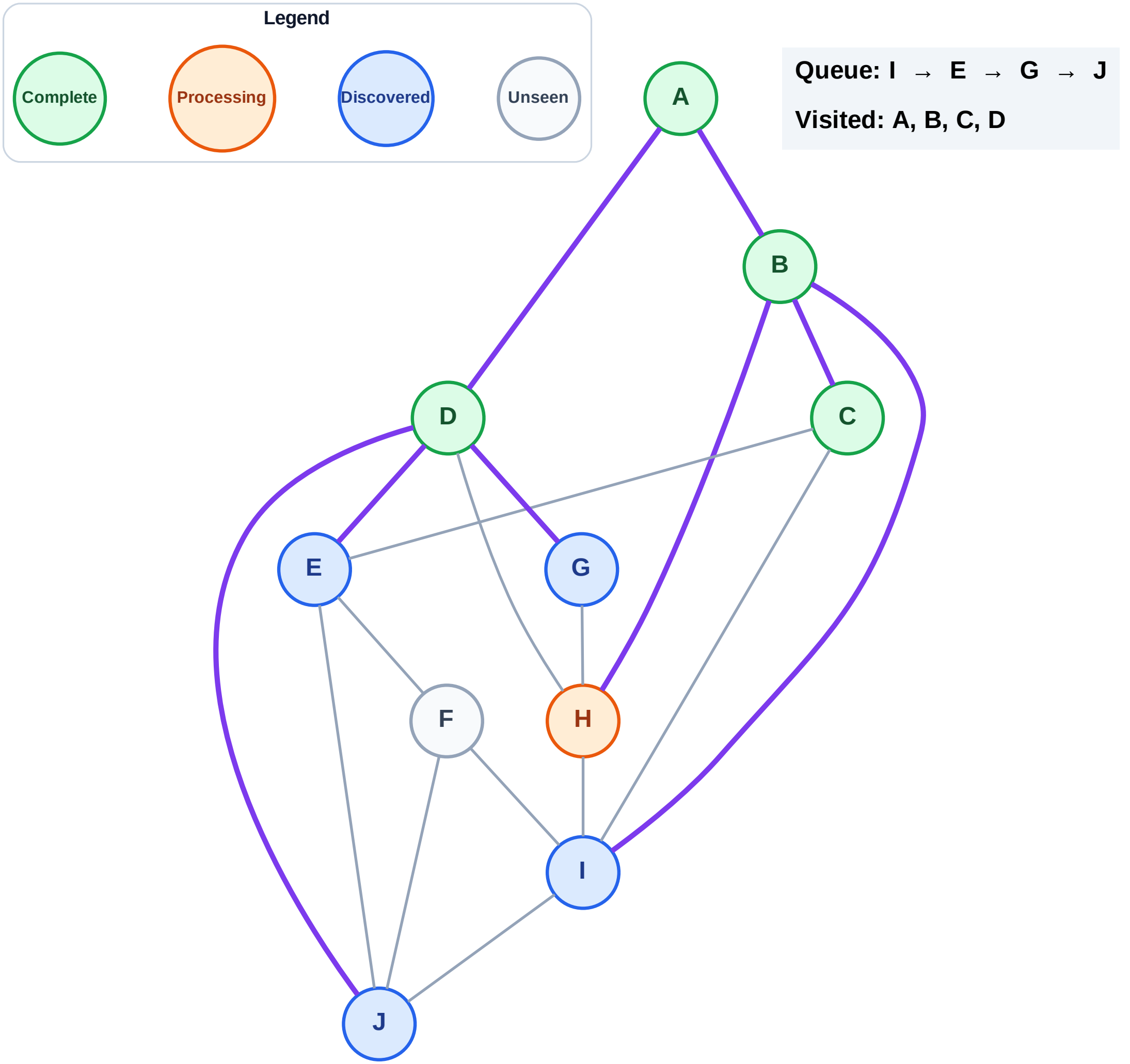


Queue: H → I → E → G → J

Visited: A, B, C, D

BFS Traversal

Step 10: Process node H

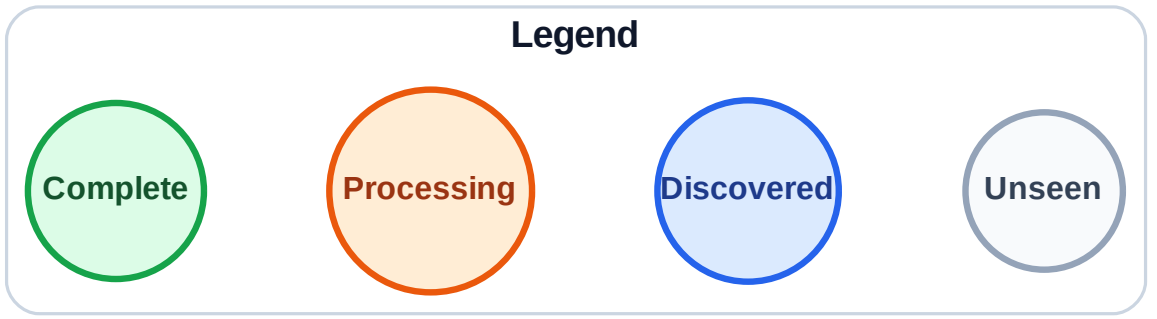


Queue: I → E → G → J

Visited: A, B, C, D

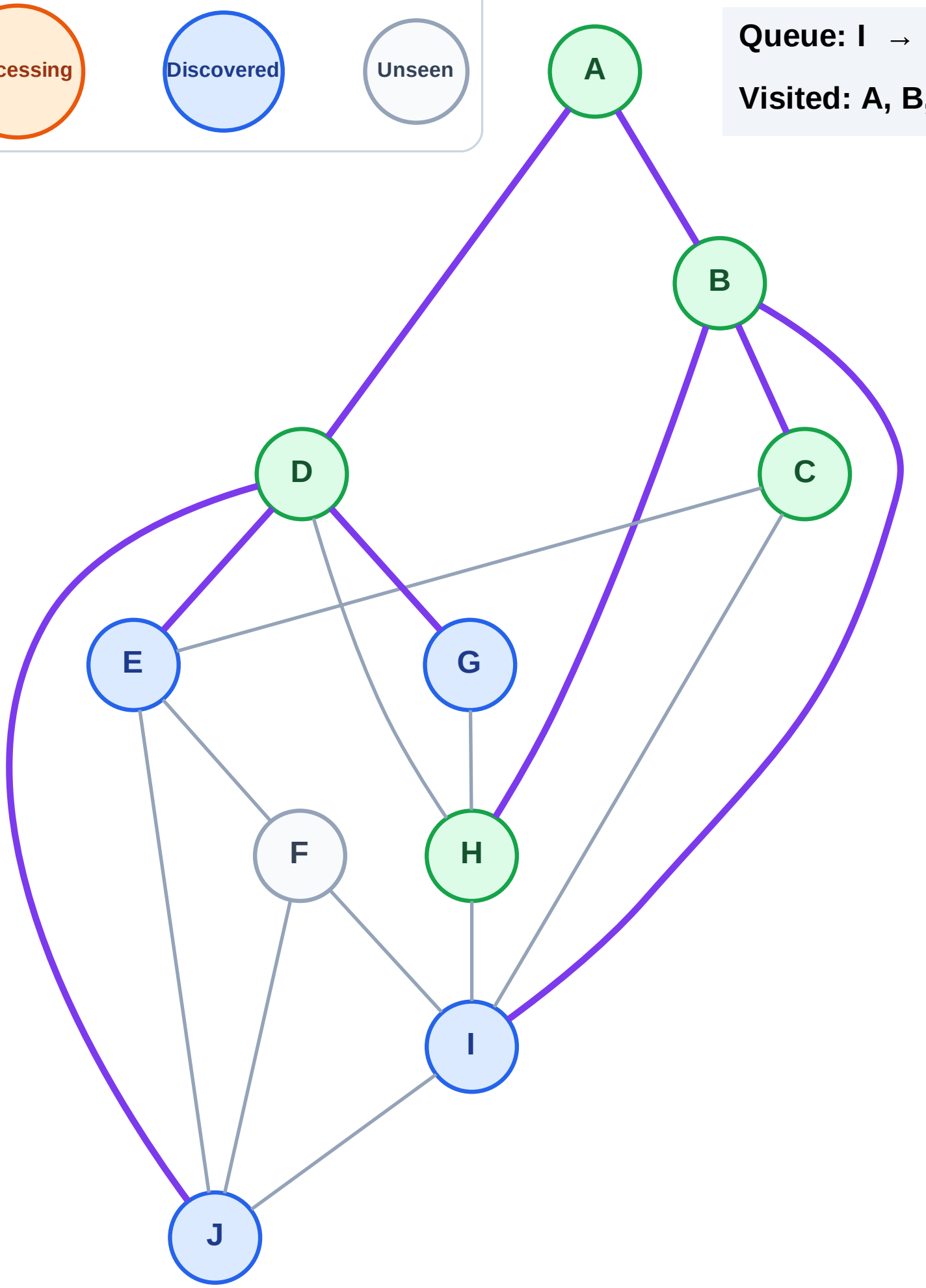
BFS Traversal

Step 11: No new neighbors from H



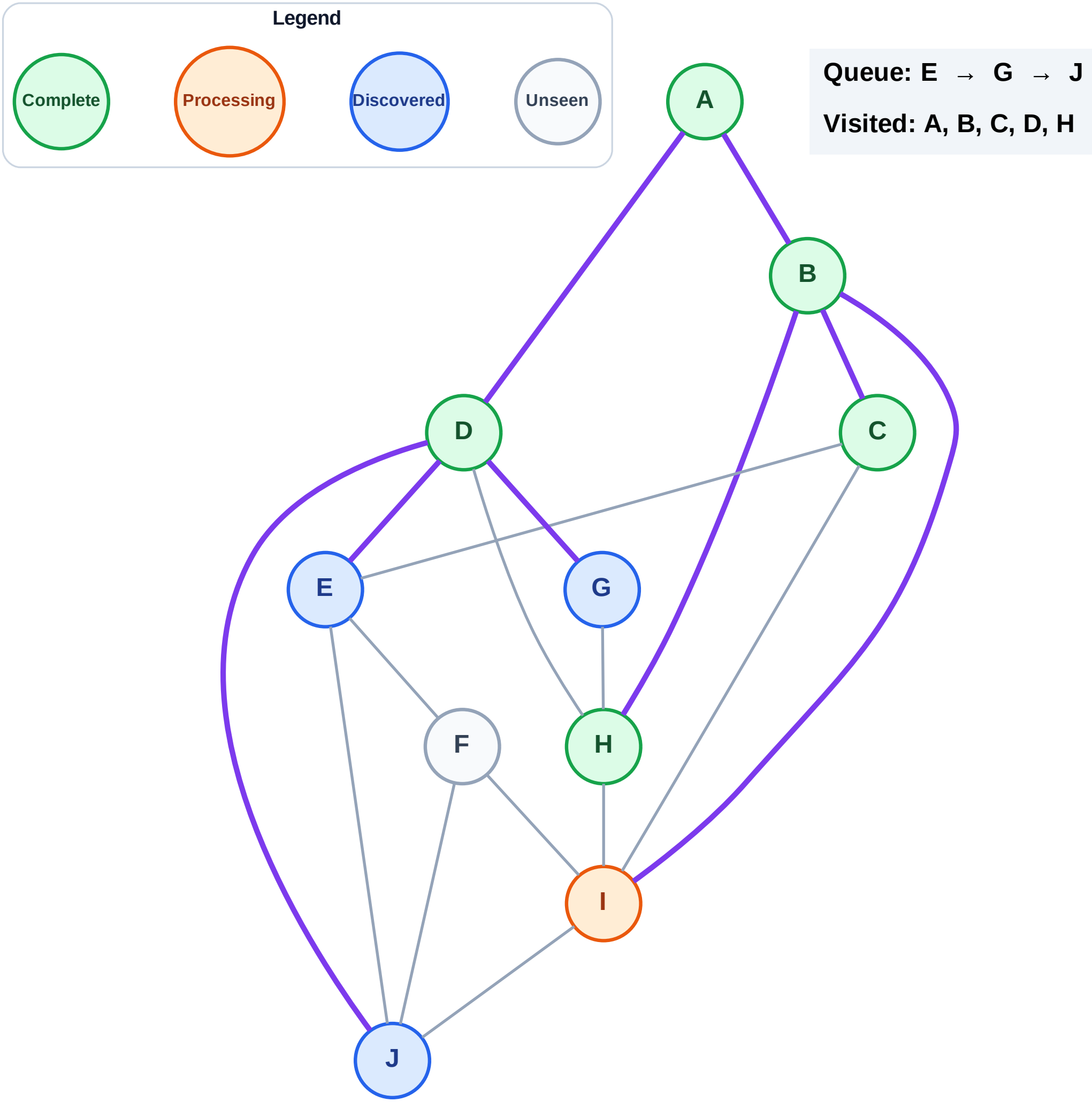
Queue: I → E → G → J

Visited: A, B, C, D, H



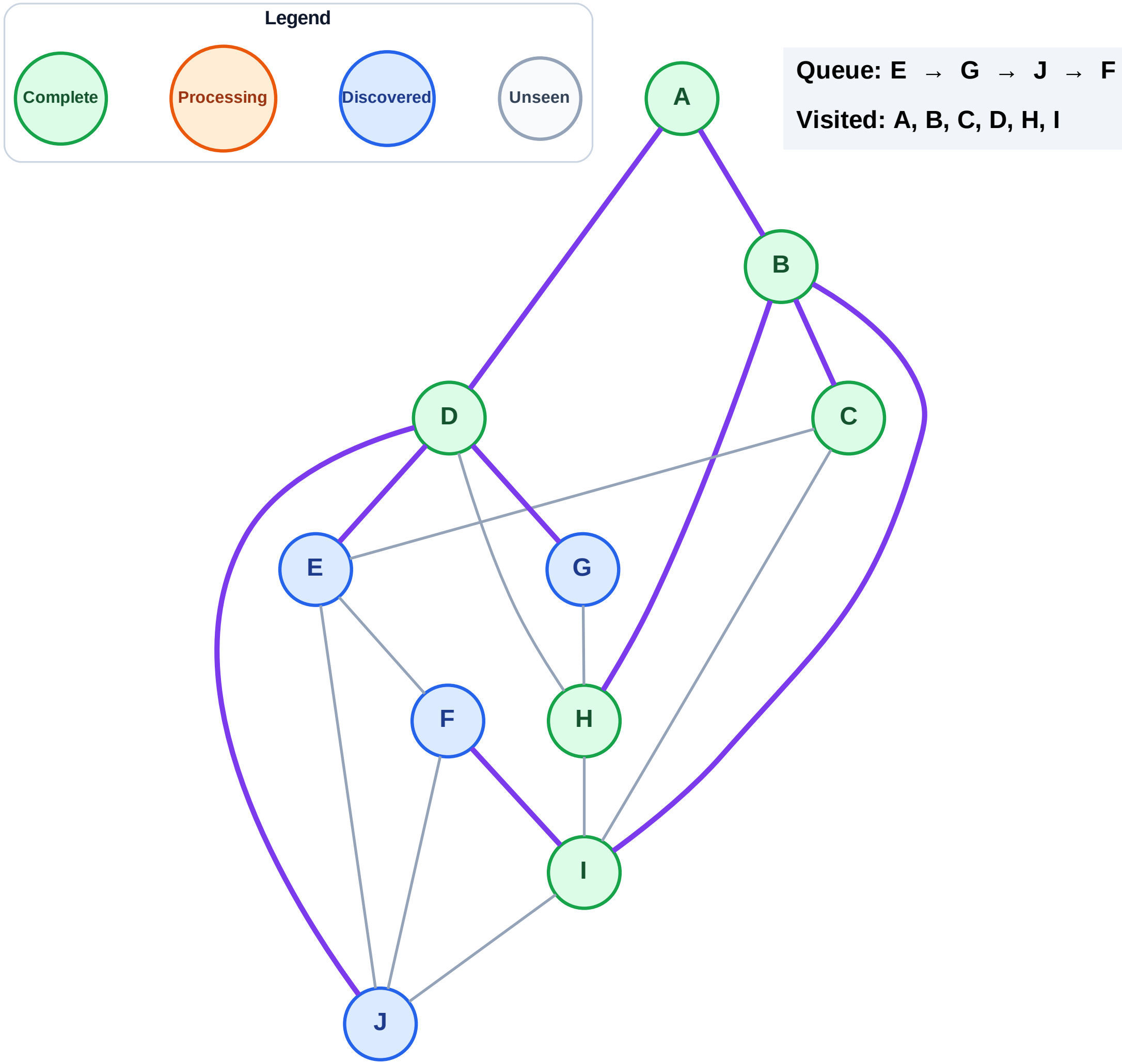
BFS Traversal

Step 12: Process node I



BFS Traversal

Step 13: Discover F from I

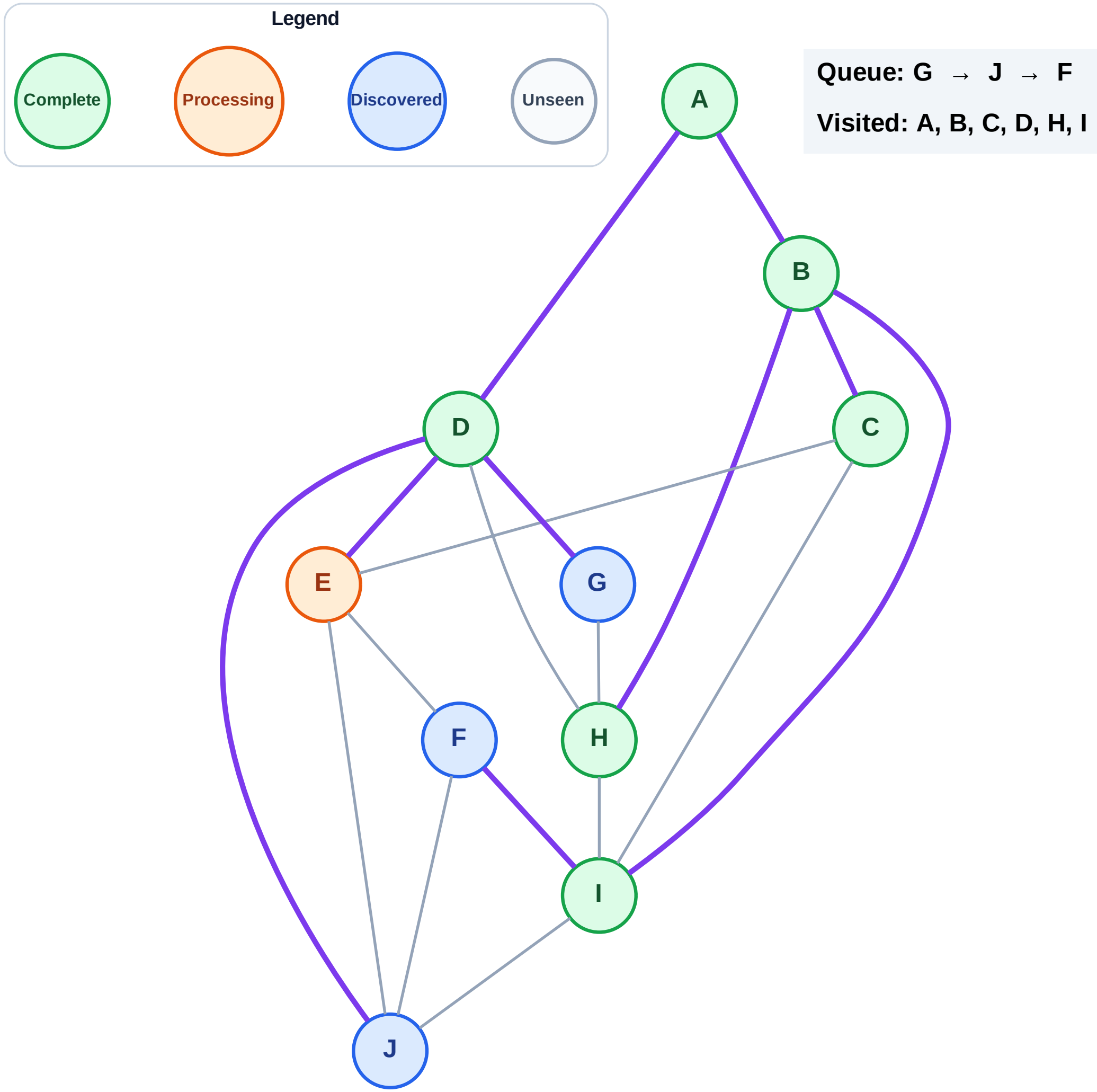


Queue: E → G → J → F

Visited: A, B, C, D, H, I

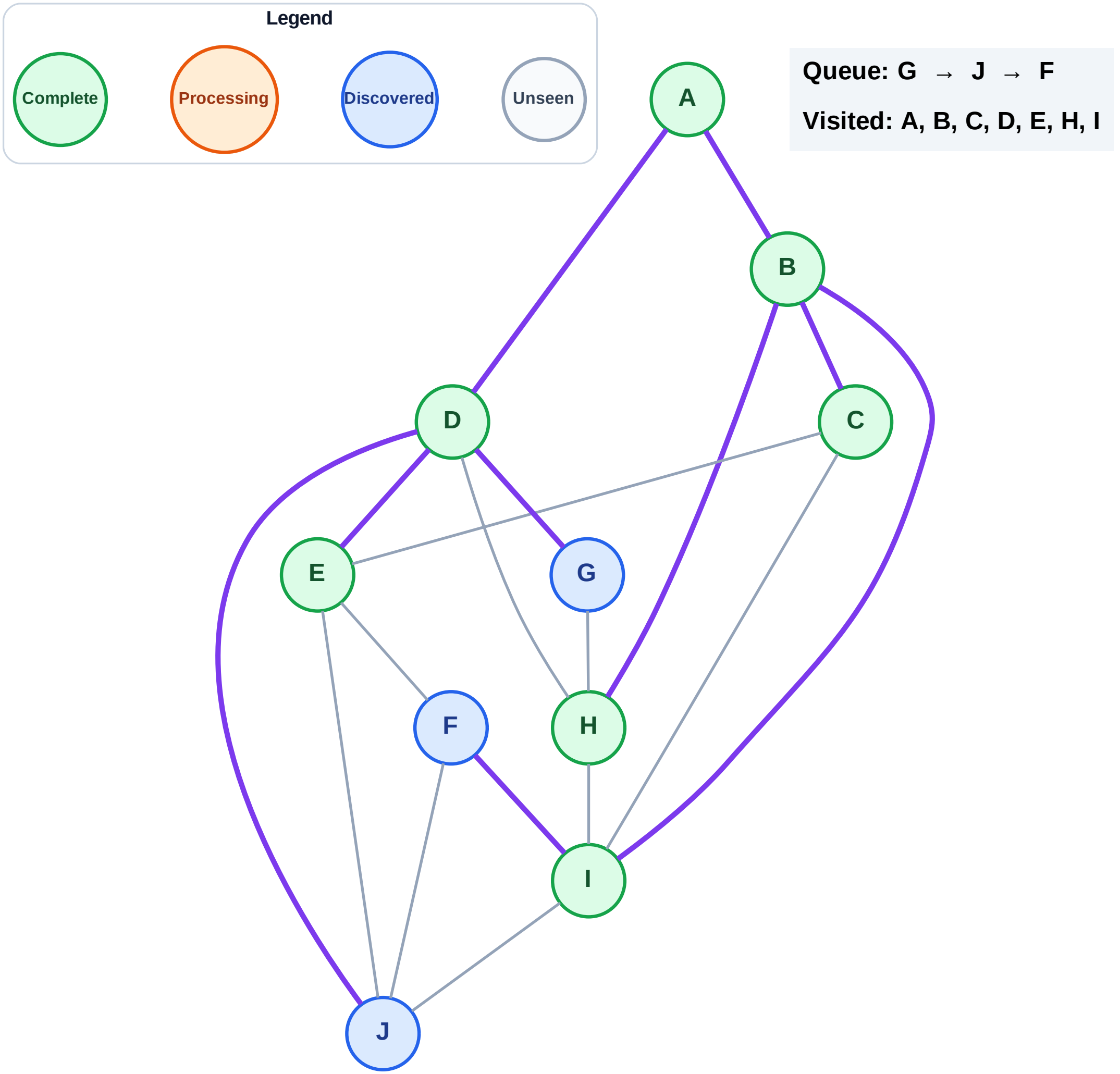
BFS Traversal

Step 14: Process node E



BFS Traversal

Step 15: No new neighbors from E

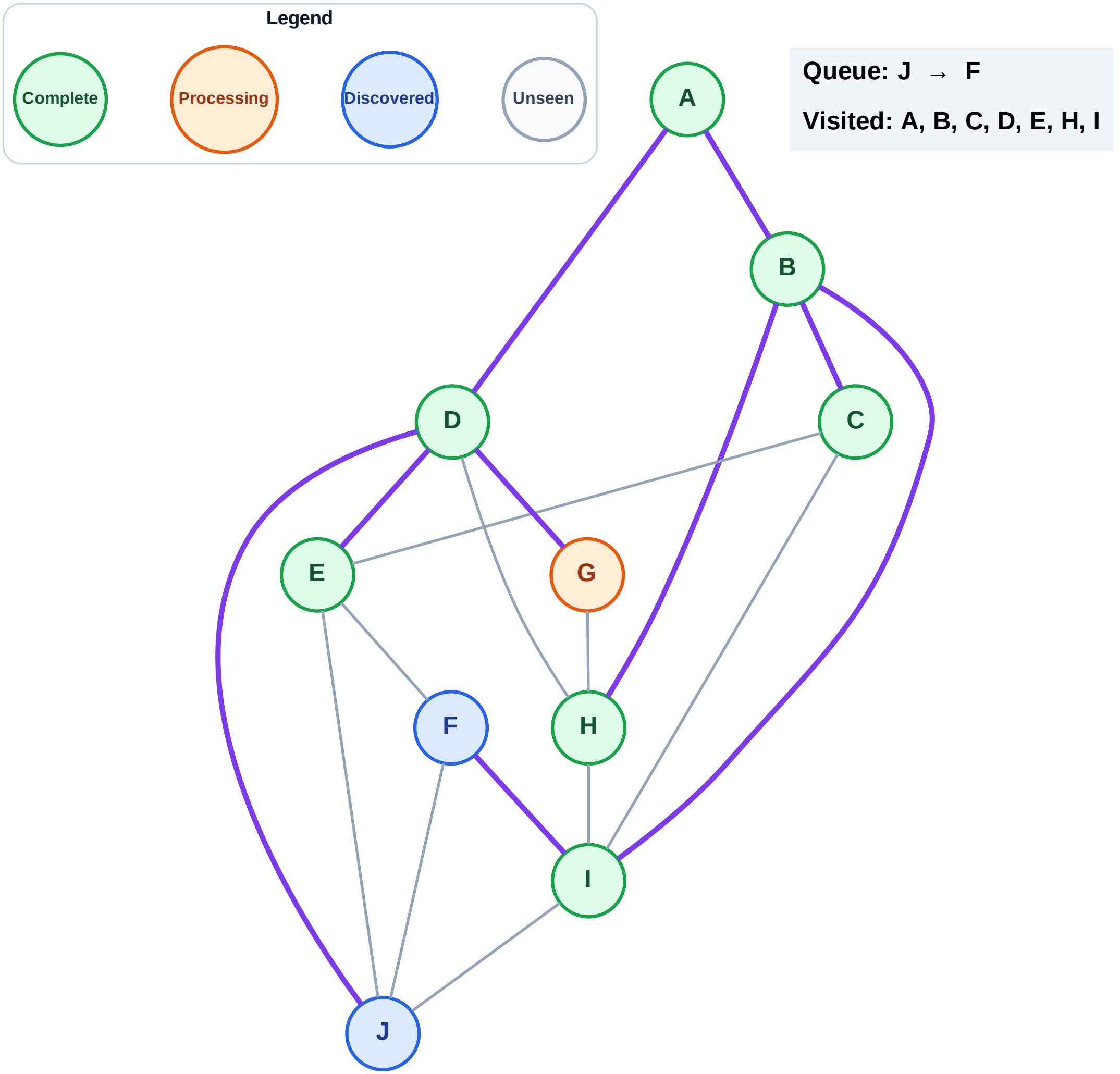


Queue: G → J → F

Visited: A, B, C, D, E, H, I

BFS Traversal

Step 16: Process node G

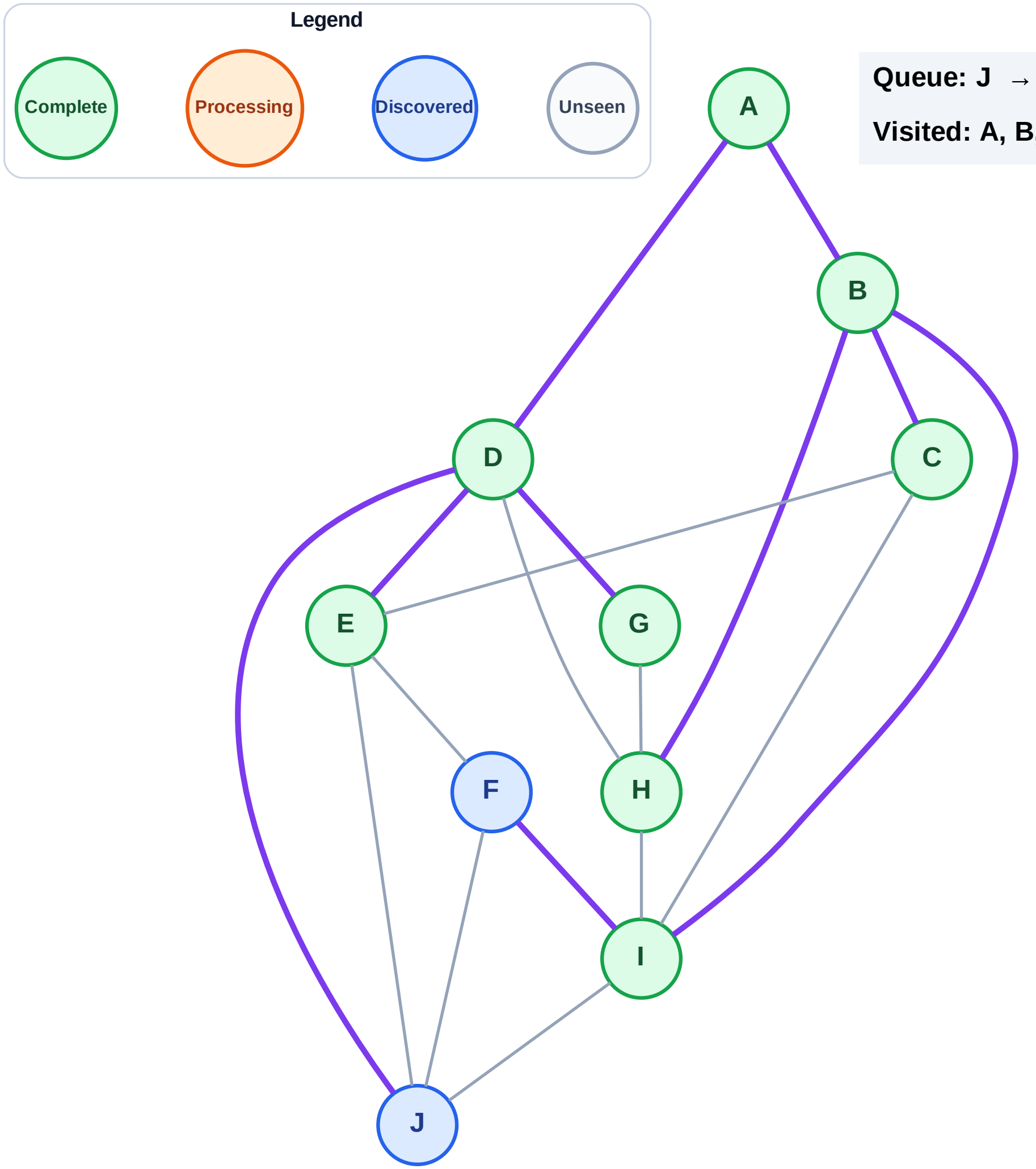


Queue: J → F

Visited: A, B, C, D, E, H, I

BFS Traversal

Step 17: No new neighbors from G

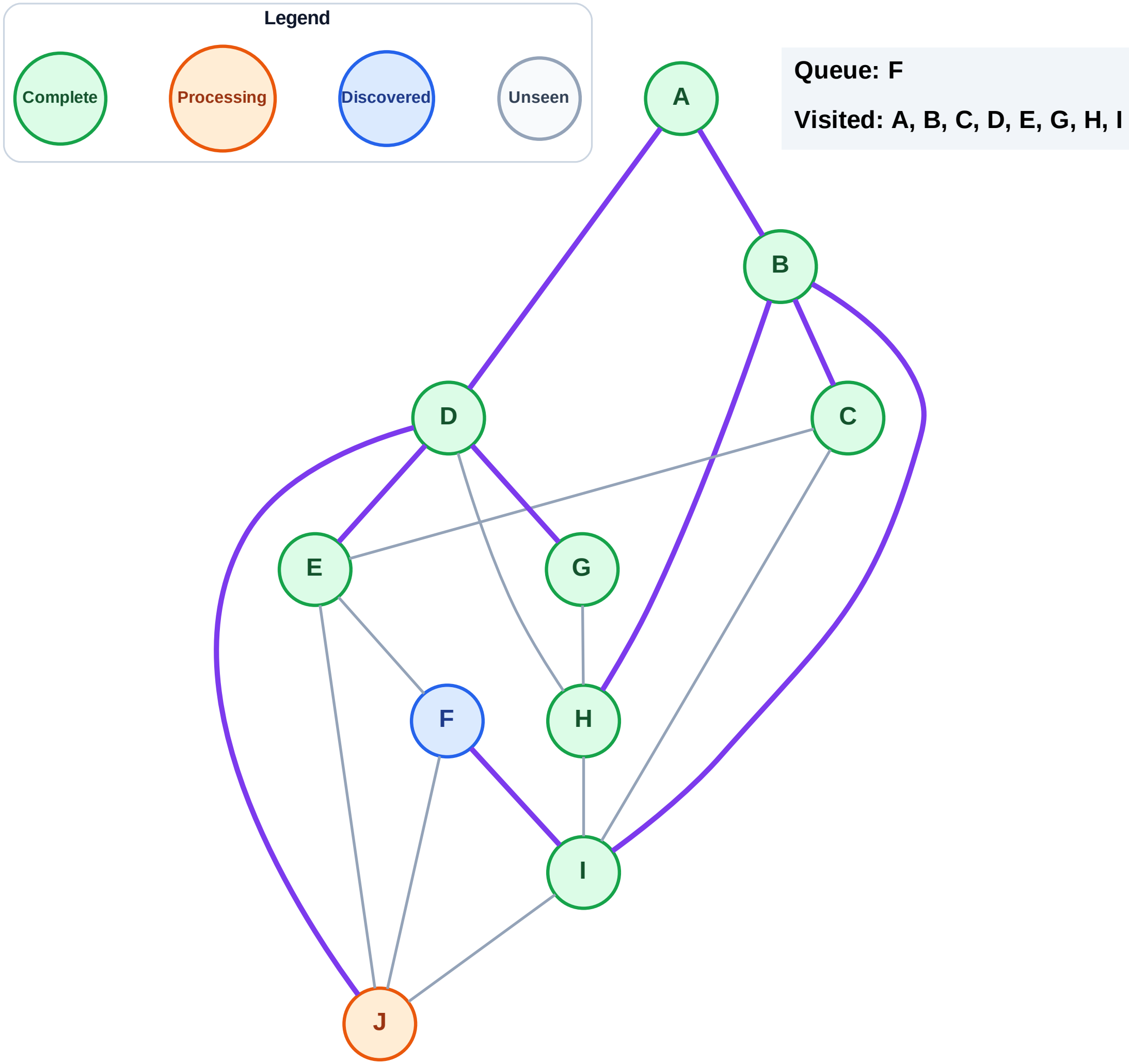


Queue: J → F

Visited: A, B, C, D, E, G, H, I

BFS Traversal

Step 18: Process node J

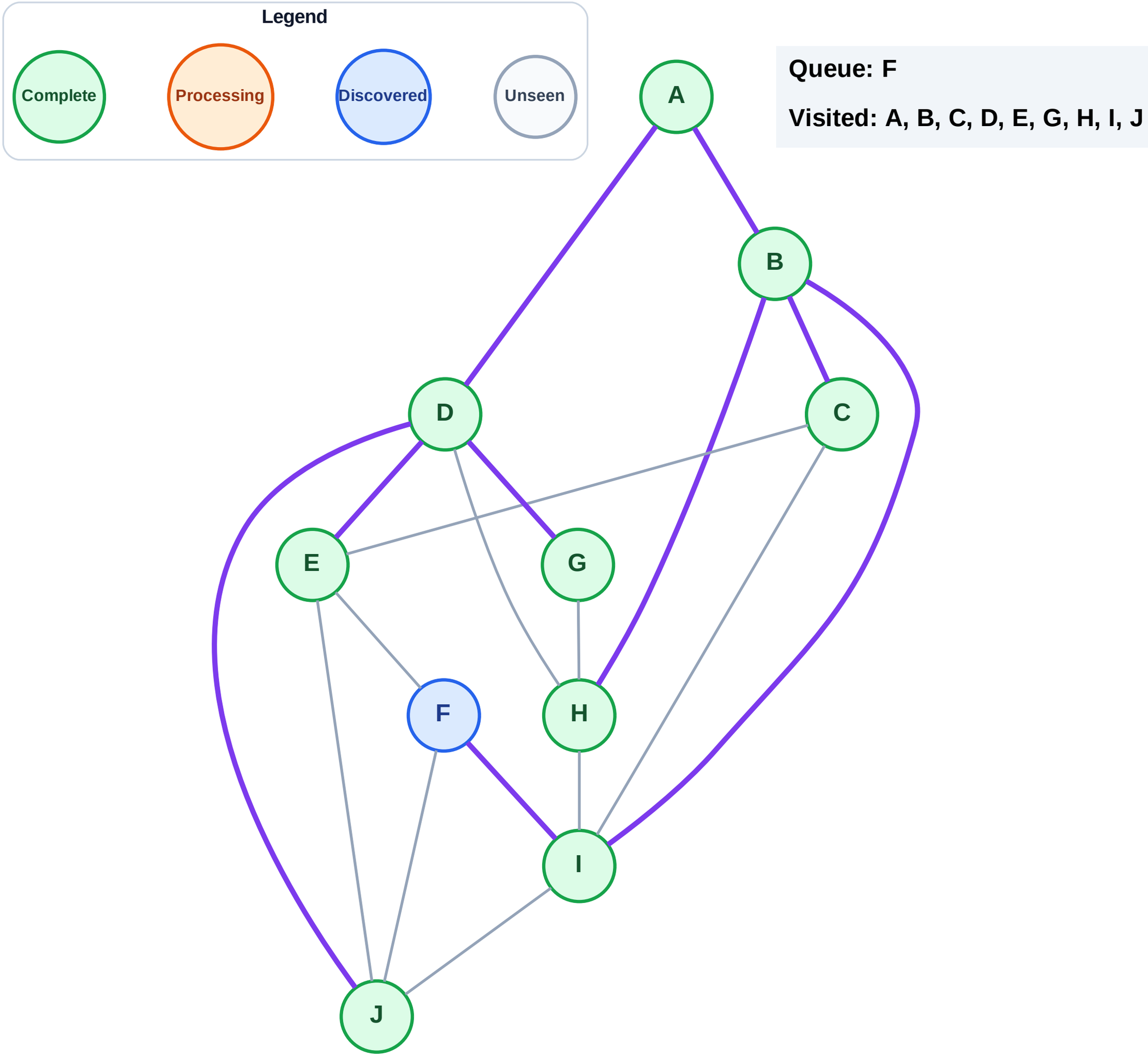


Queue: F

Visited: A, B, C, D, E, G, H, I

BFS Traversal

Step 19: No new neighbors from J

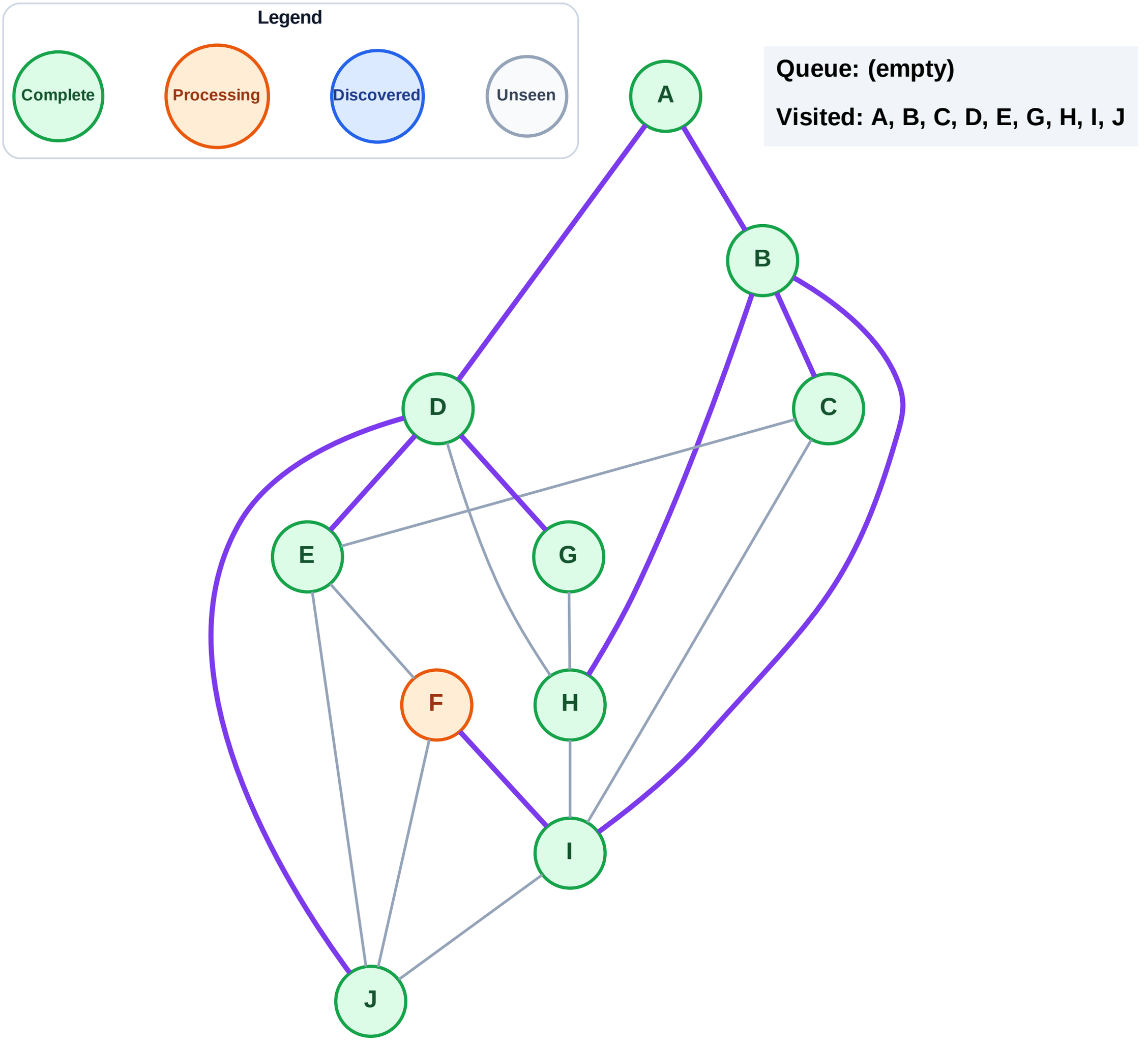


Queue: F

Visited: A, B, C, D, E, G, H, I, J

BFS Traversal

Step 20: Process node F

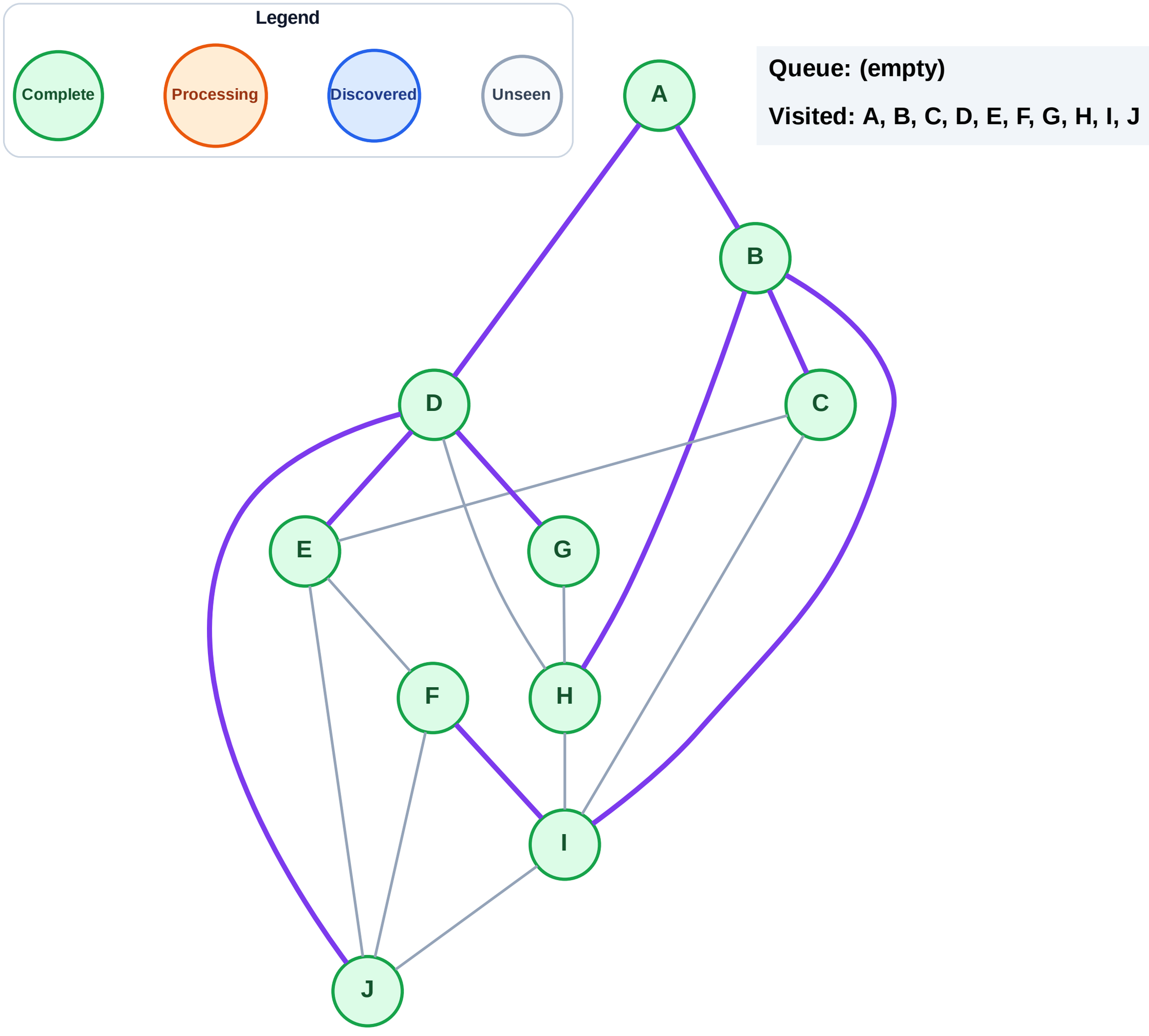


Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J

BFS Traversal

Step 21: No new neighbors from F

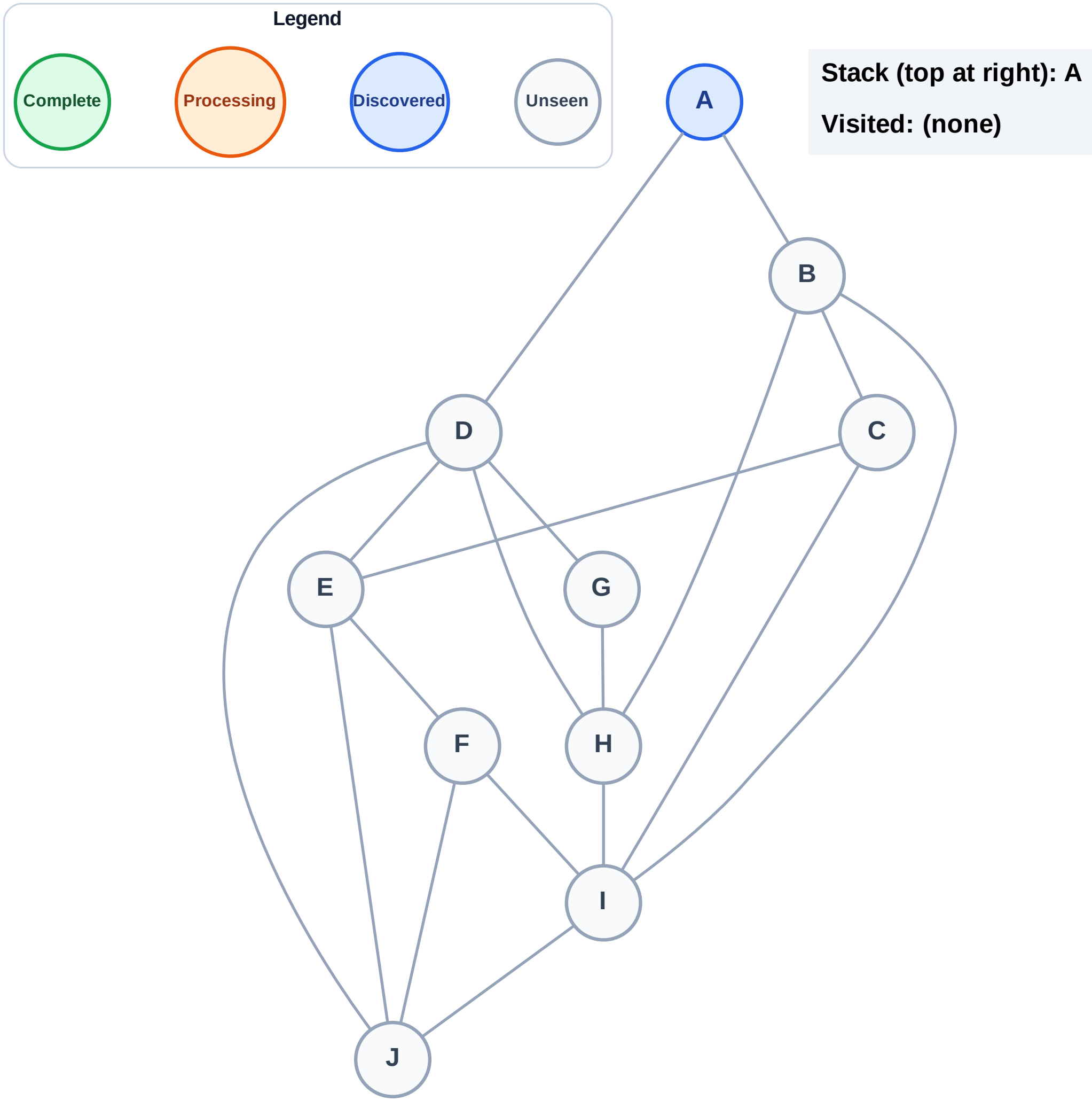


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at A

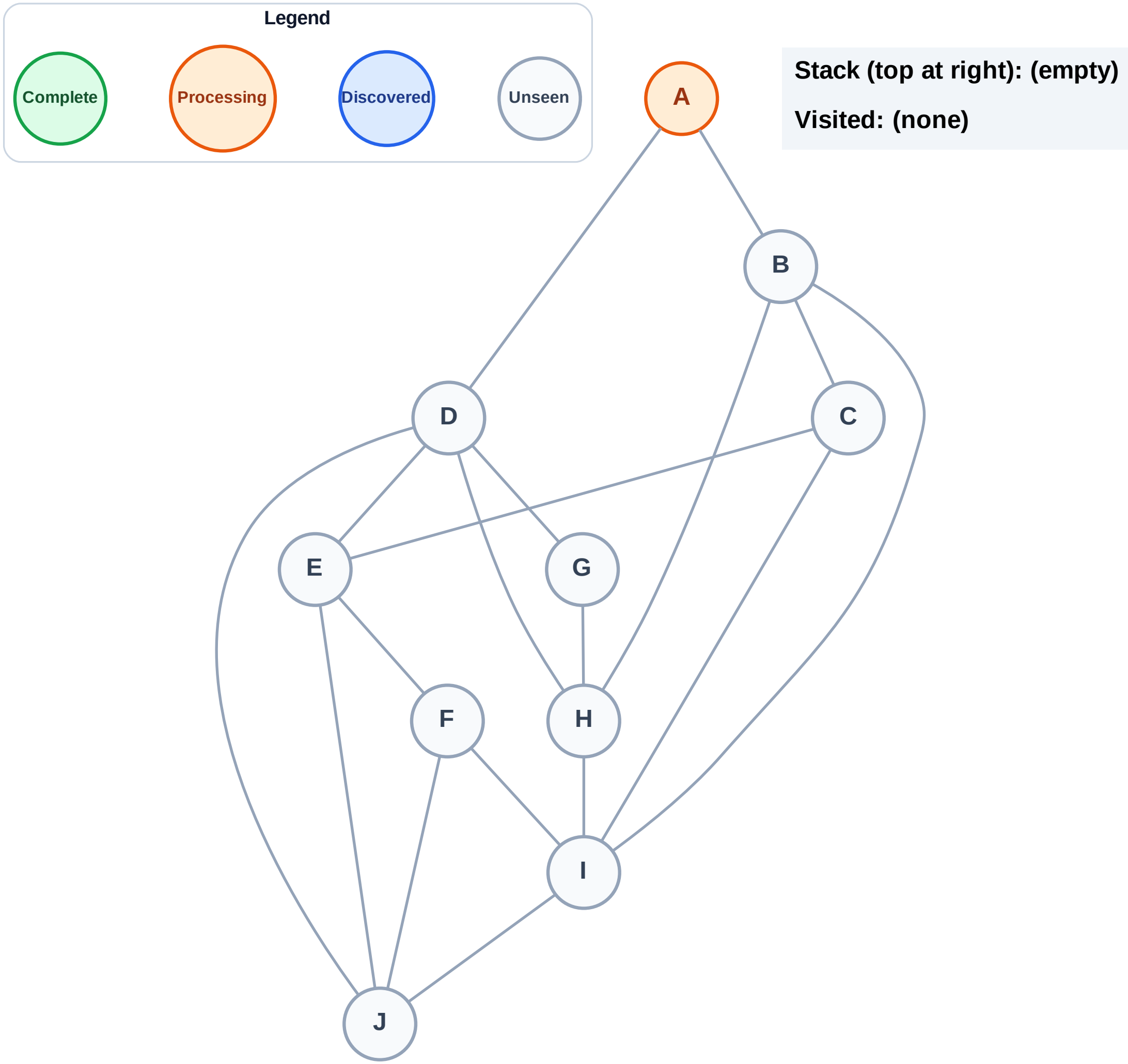


Stack (top at right): A

Visited: (none)

DFS Traversal

Step 2: Process node A

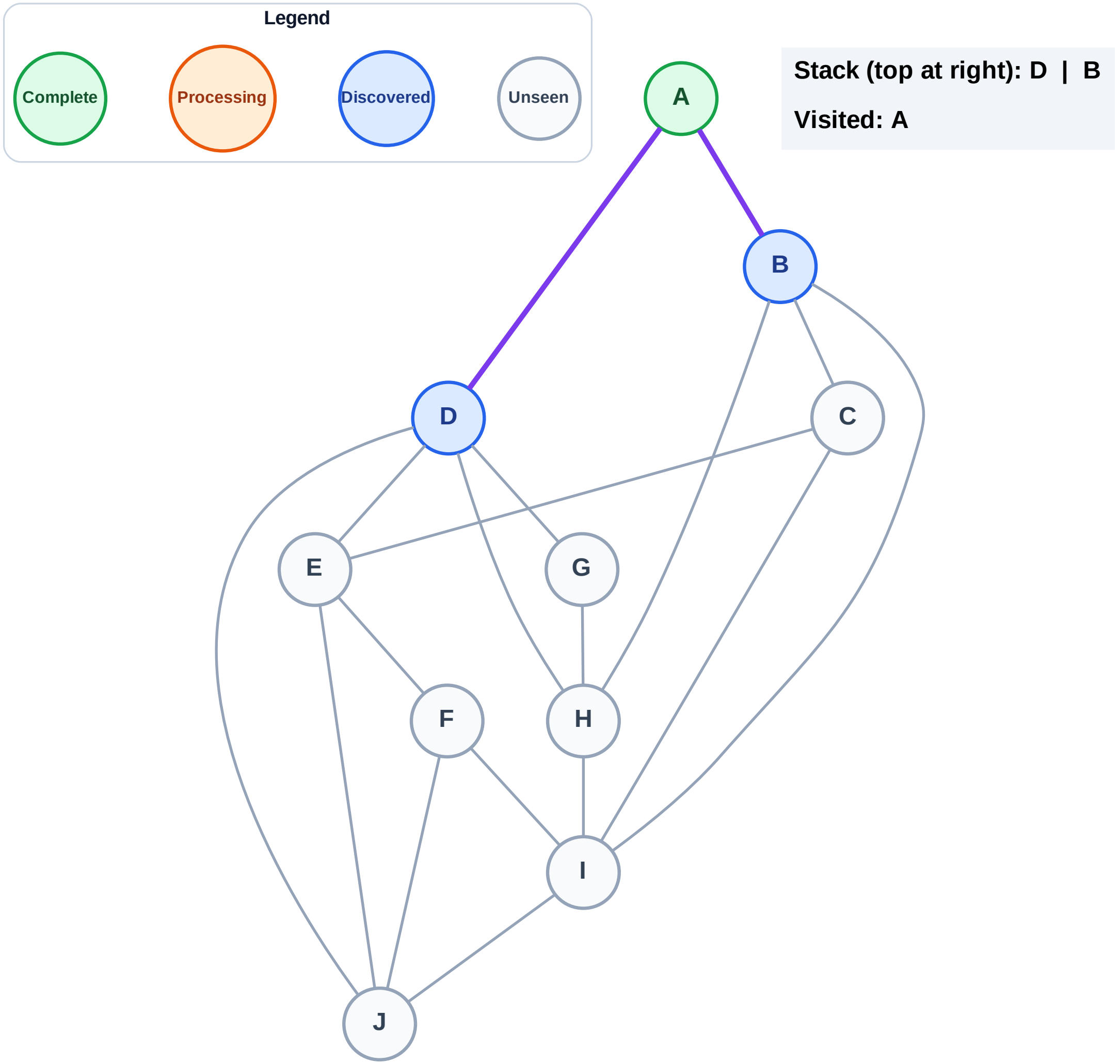


Stack (top at right): (empty)

Visited: (none)

DFS Traversal

Step 3: Discover D, B from A

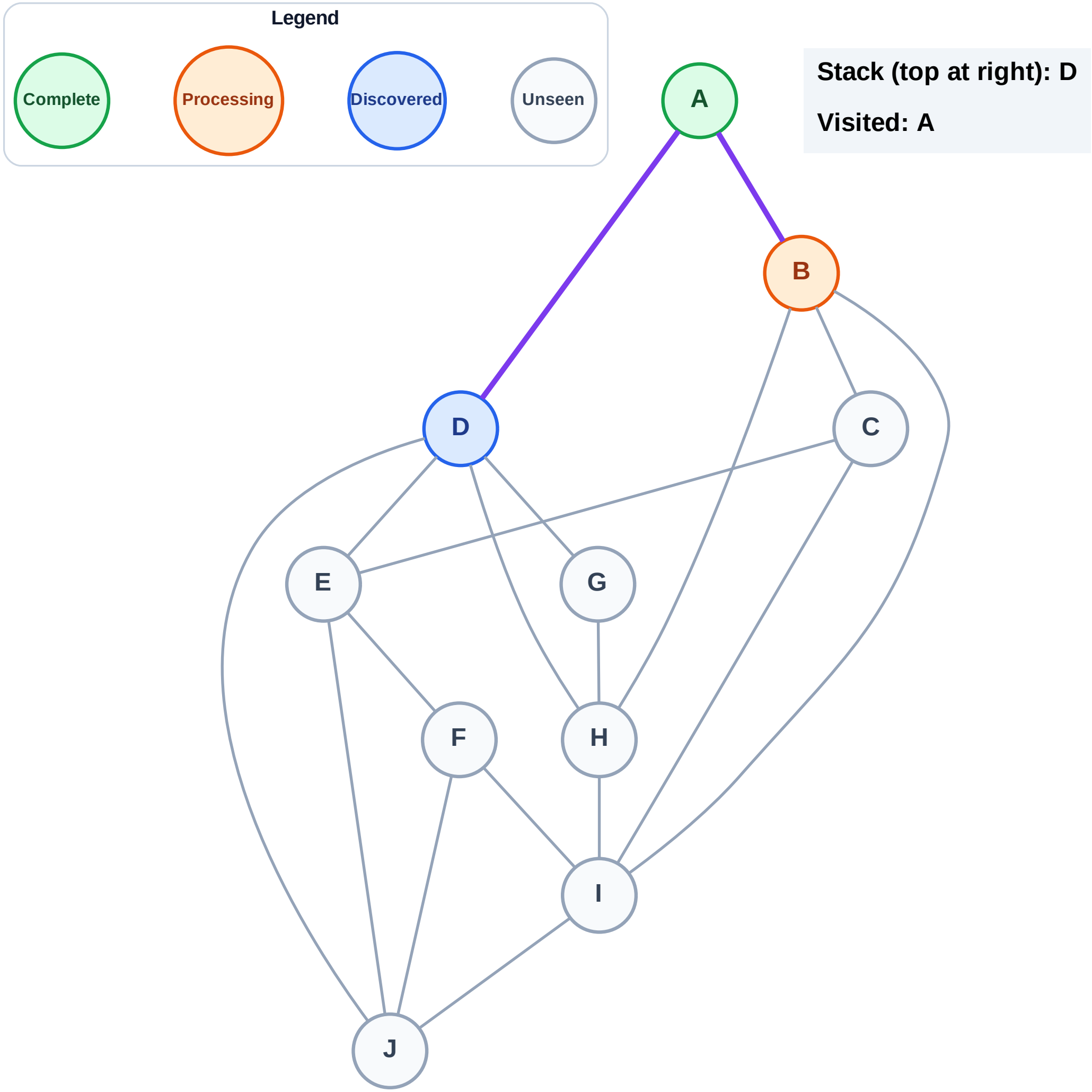


Stack (top at right): D | B

Visited: A

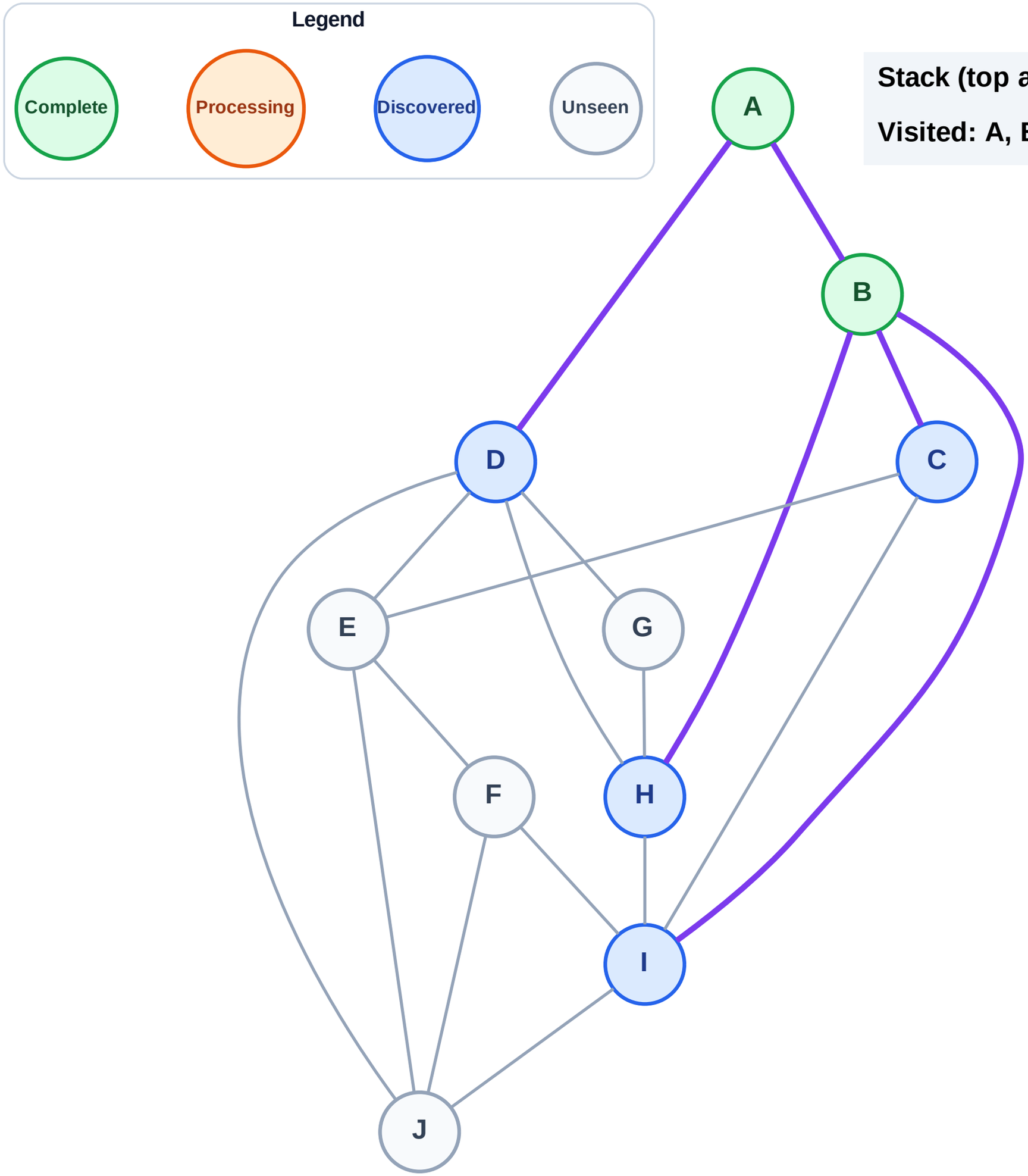
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover I, H, C from B

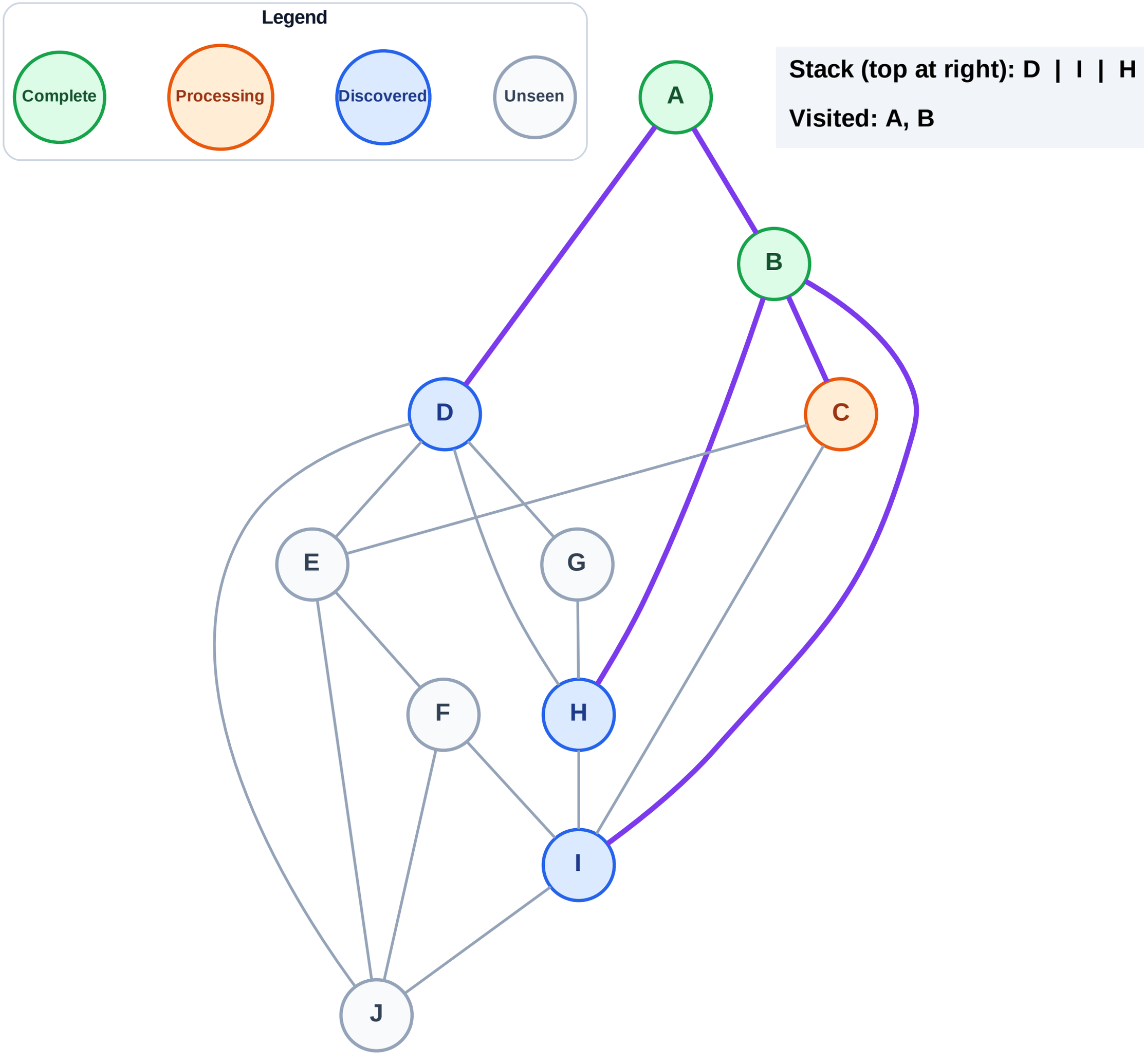


Stack (top at right): D | I | H | C

Visited: A, B

DFS Traversal

Step 6: Process node C

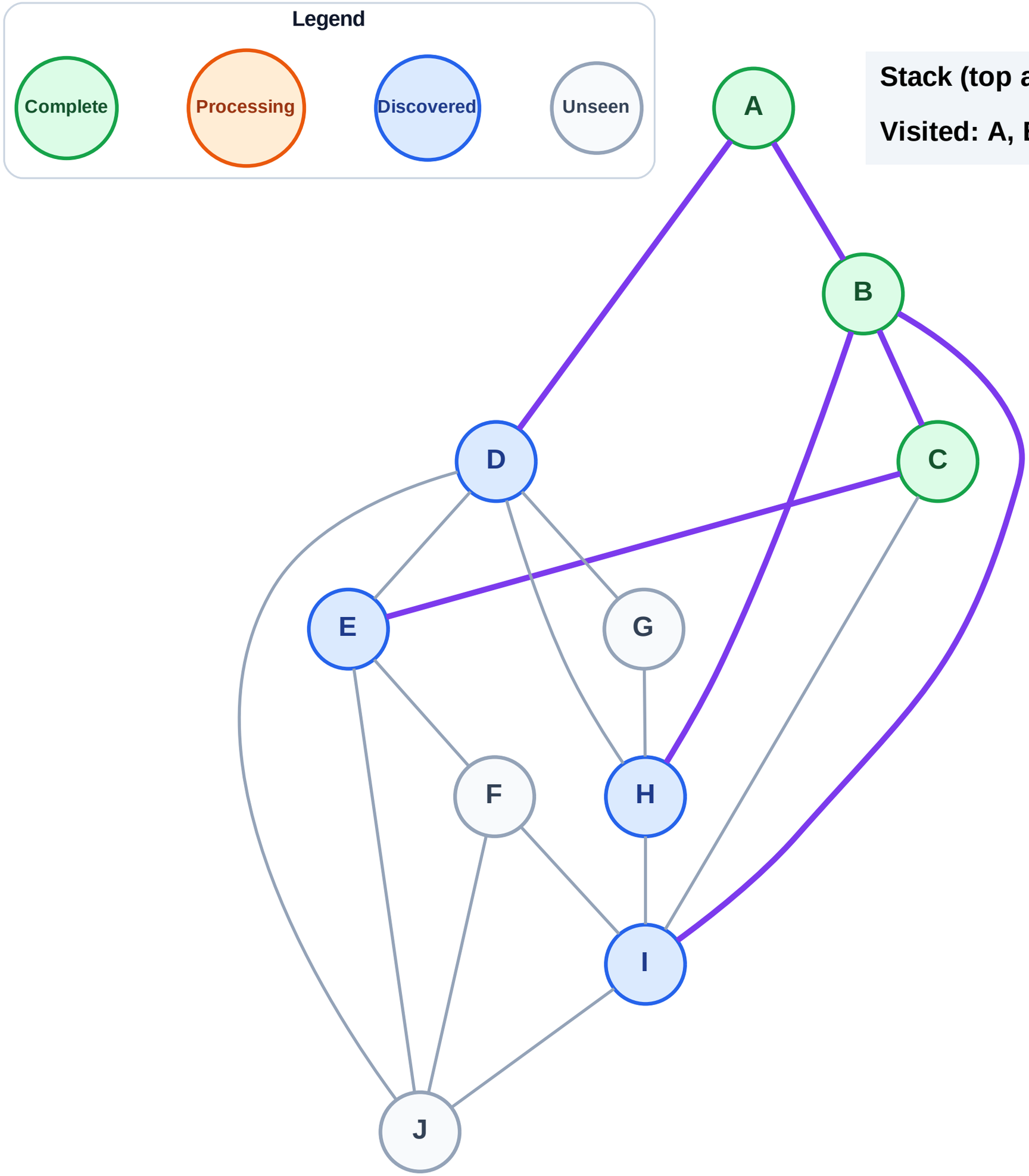


Stack (top at right): D | I | H

Visited: A, B

DFS Traversal

Step 7: Discover E from C

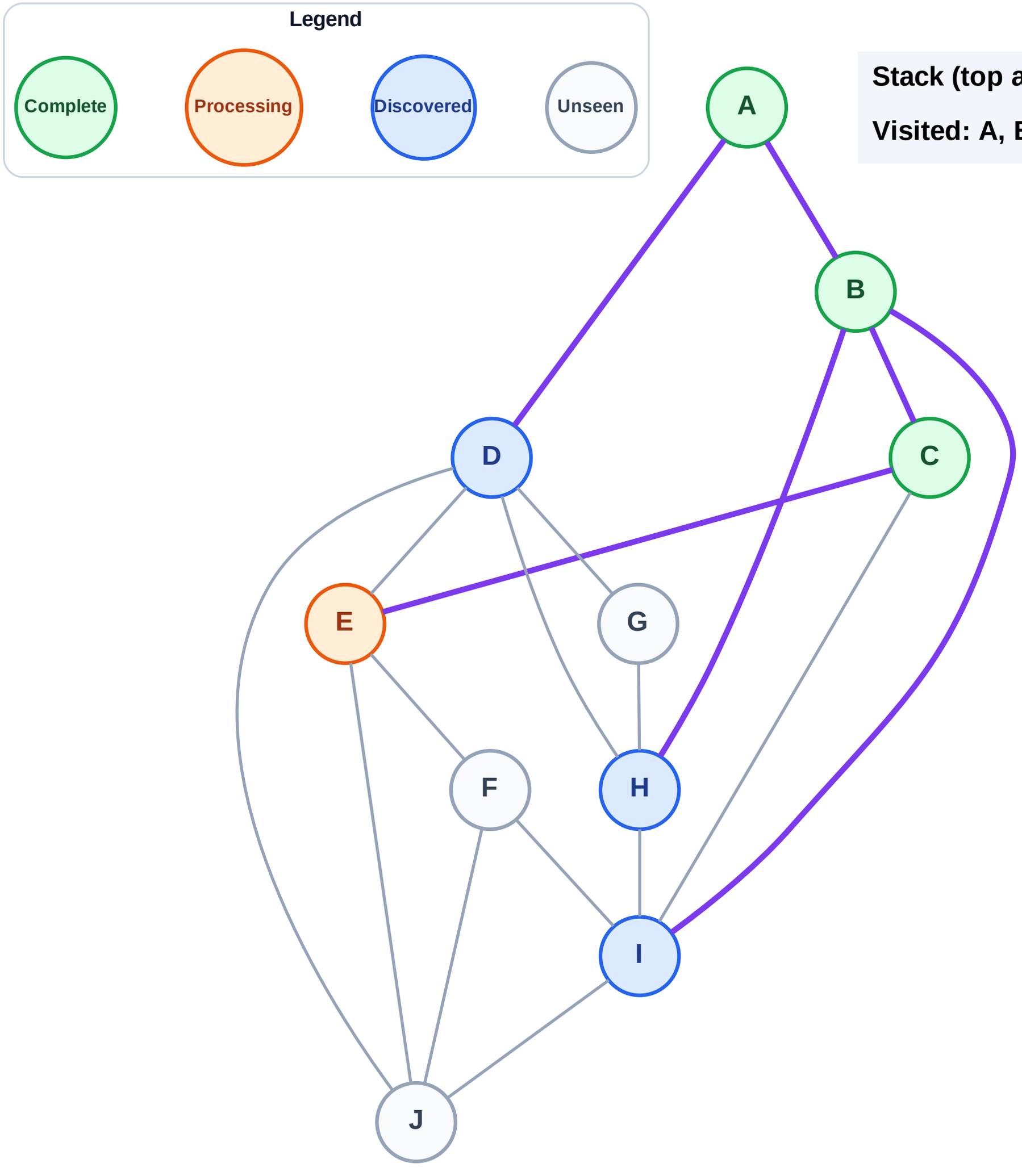


Stack (top at right): D | I | H | E

Visited: A, B, C

DFS Traversal

Step 8: Process node E

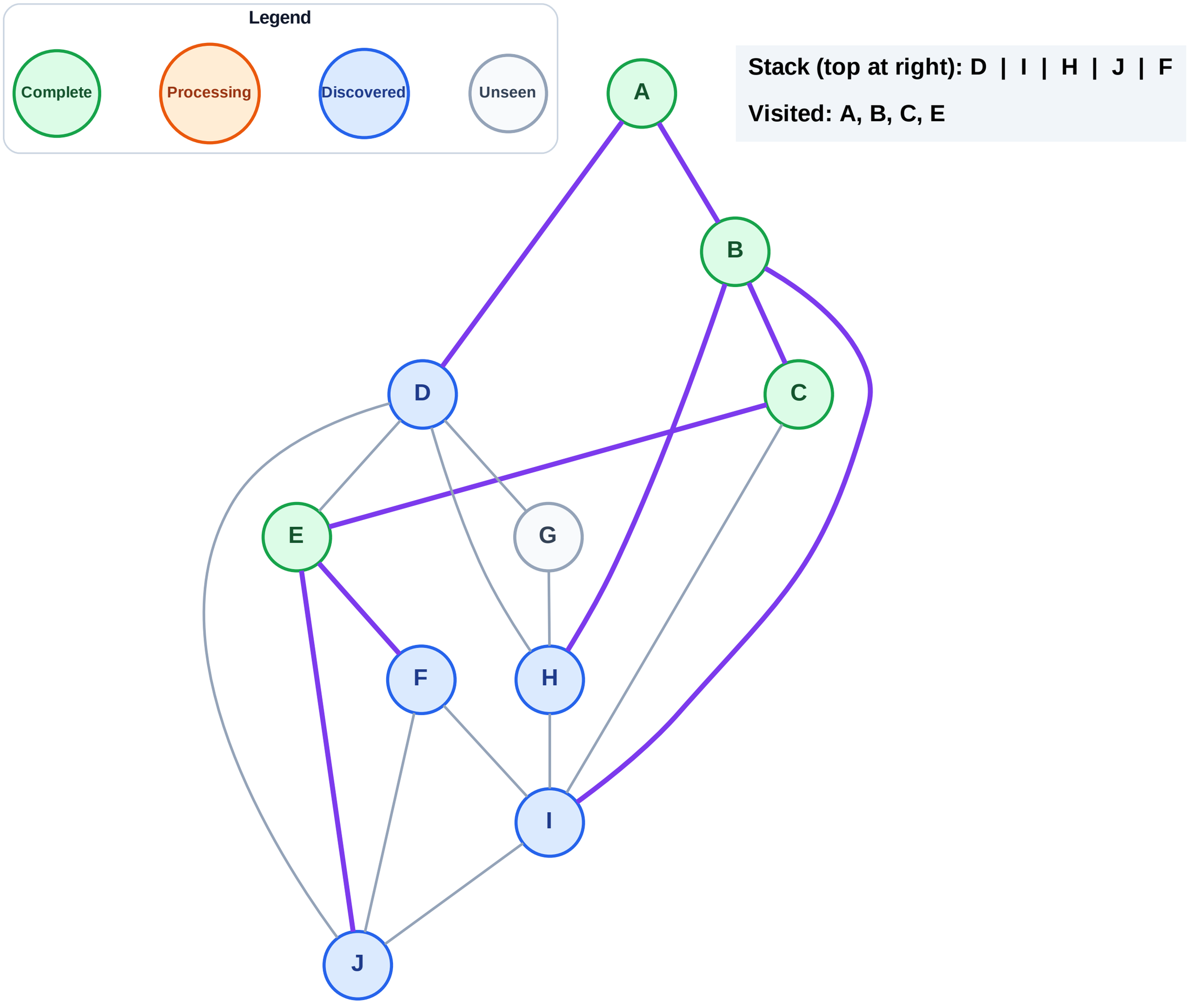


Stack (top at right): D | I | H

Visited: A, B, C

DFS Traversal

Step 9: Discover J, F from E

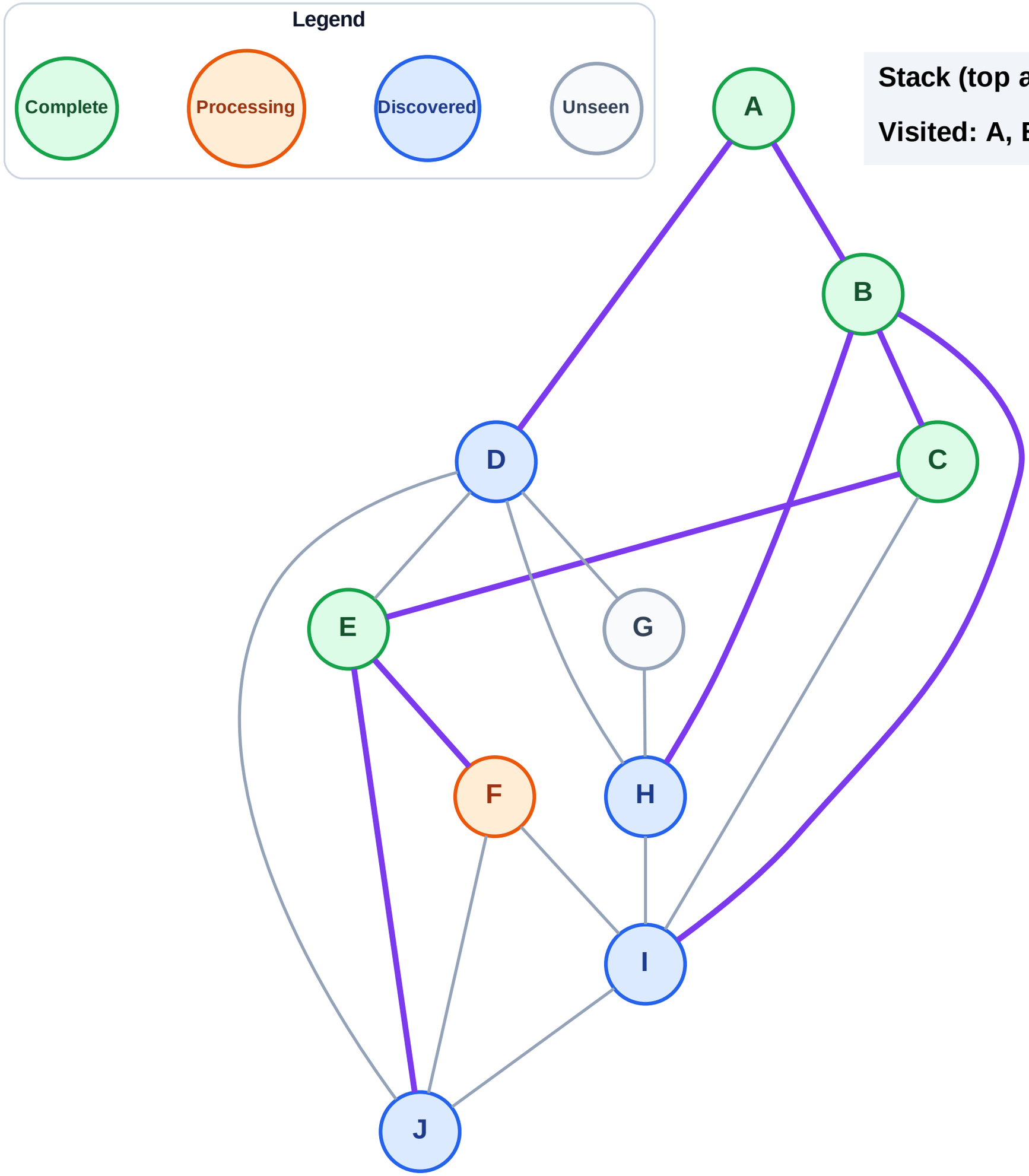


Stack (top at right): D | I | H | J | F

Visited: A, B, C, E

DFS Traversal

Step 10: Process node F

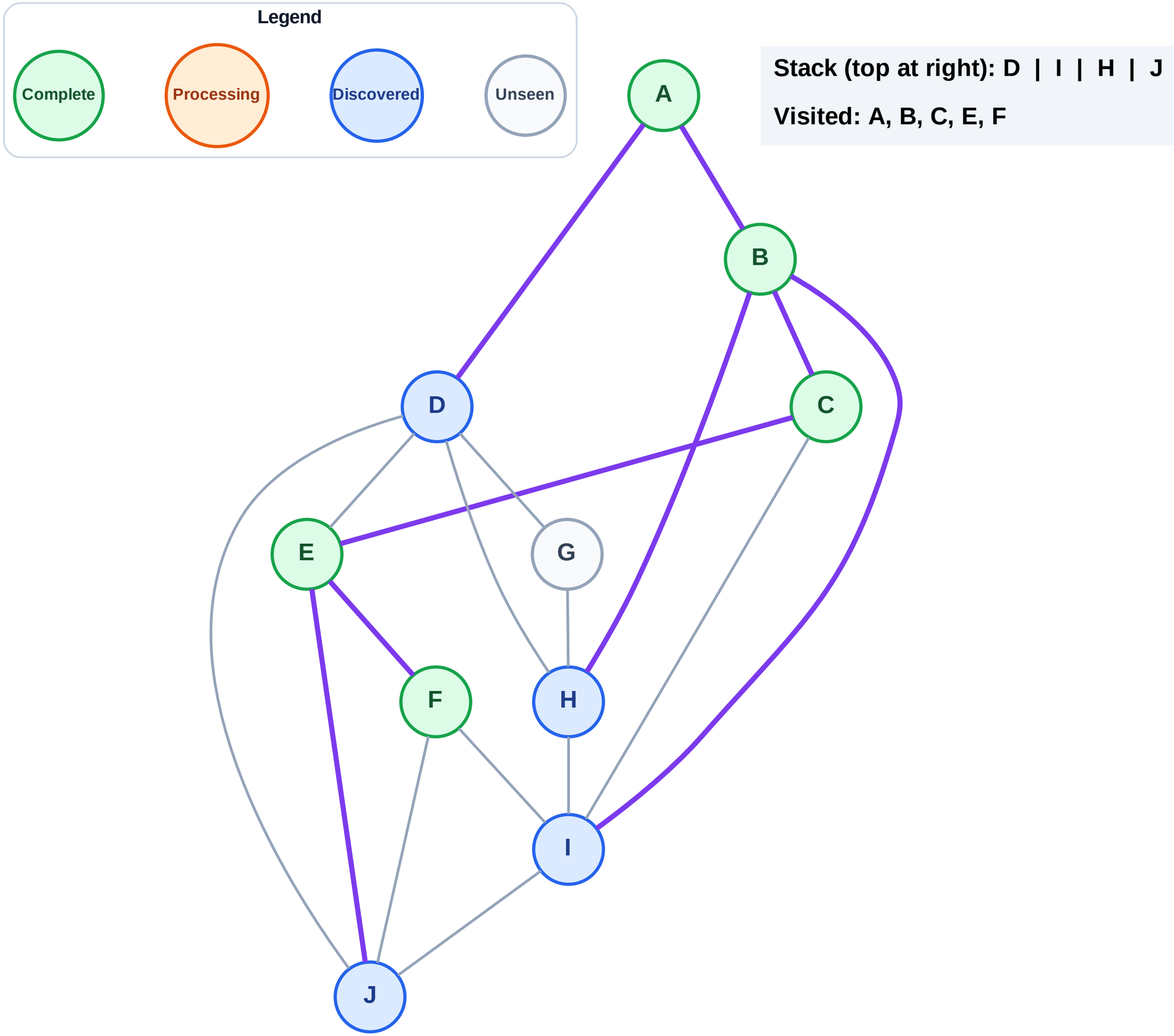


Stack (top at right): D | I | H | J

Visited: A, B, C, E

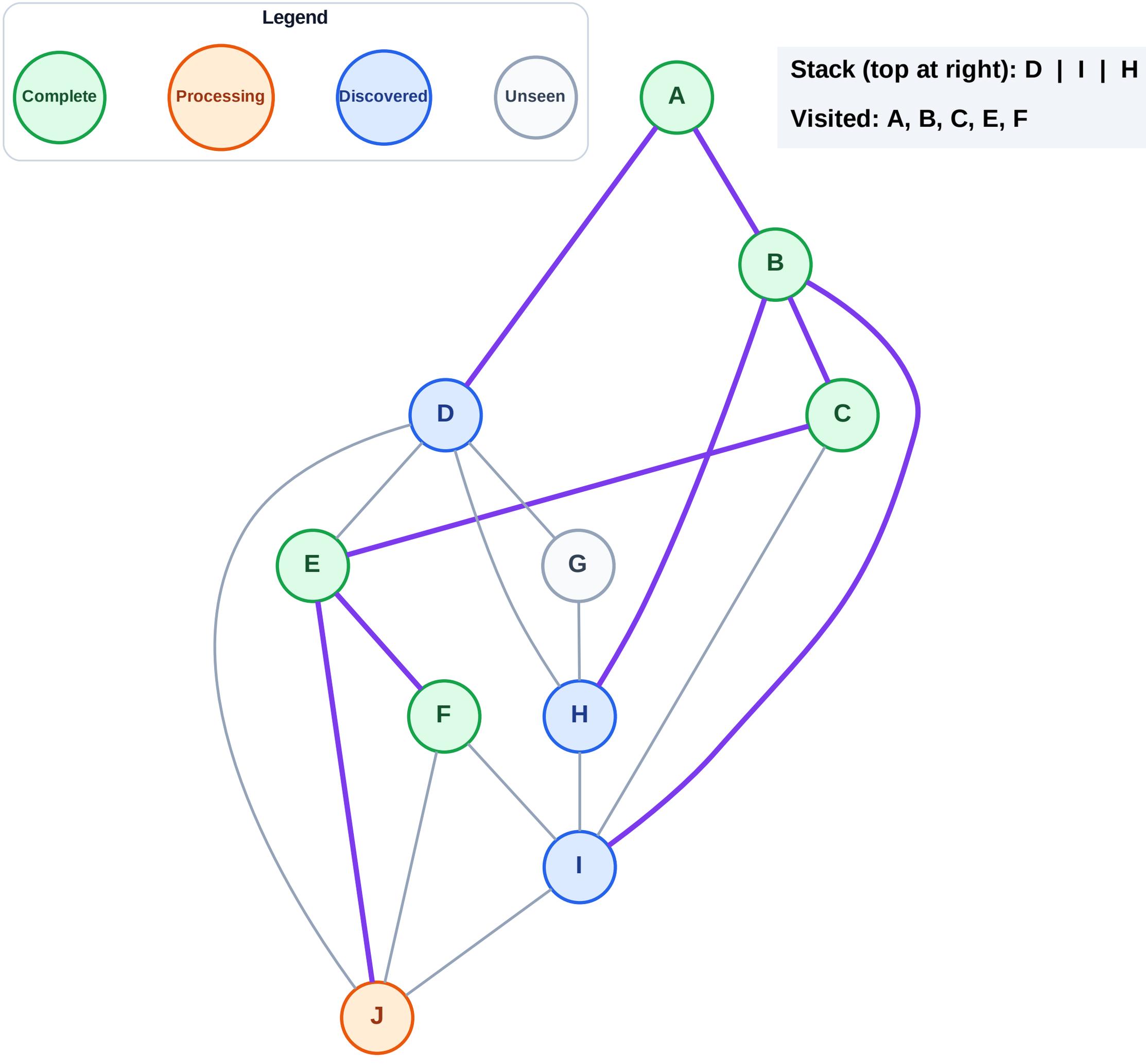
DFS Traversal

Step 11: No new neighbors from F



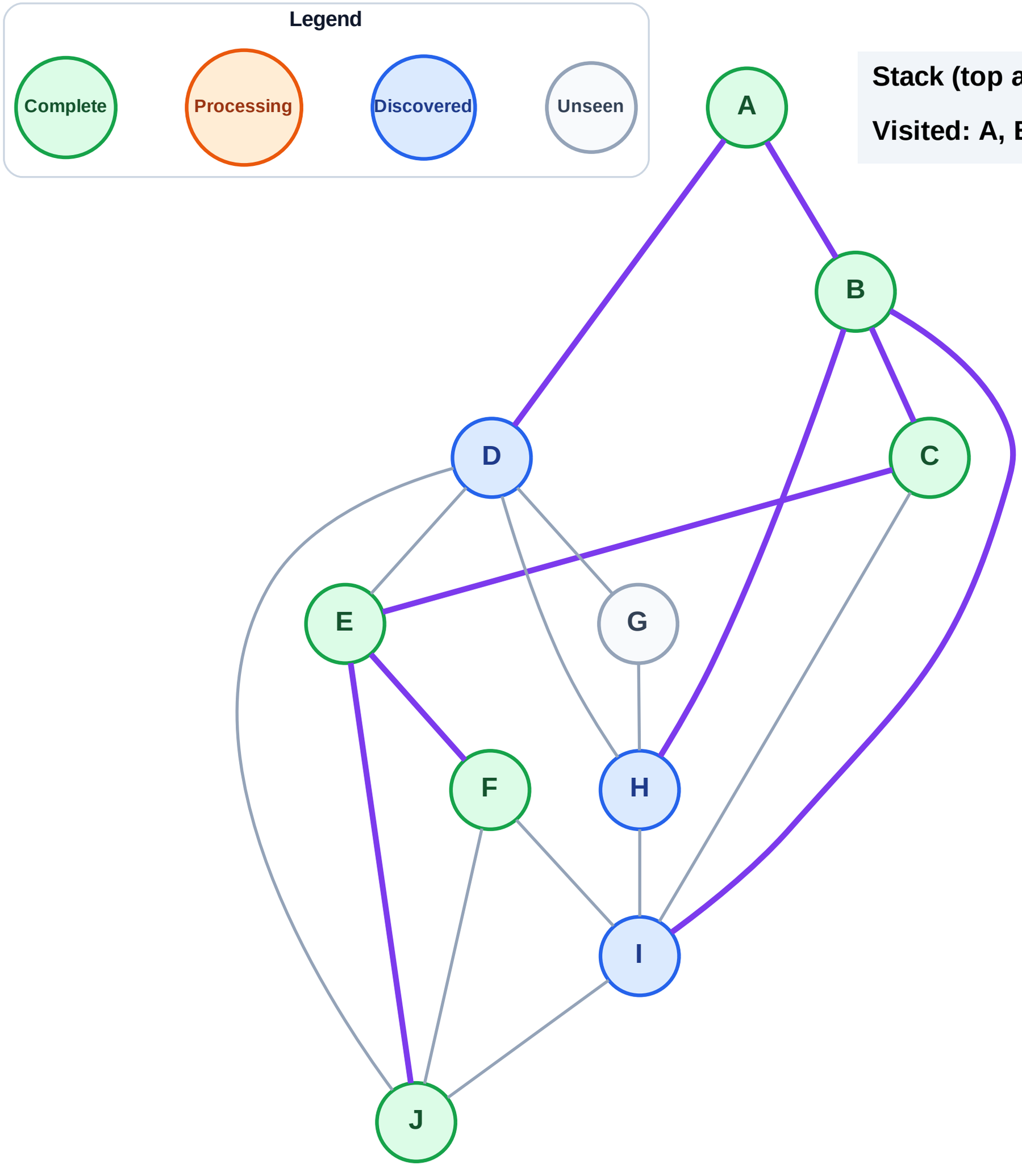
DFS Traversal

Step 12: Process node J



DFS Traversal

Step 13: No new neighbors from J

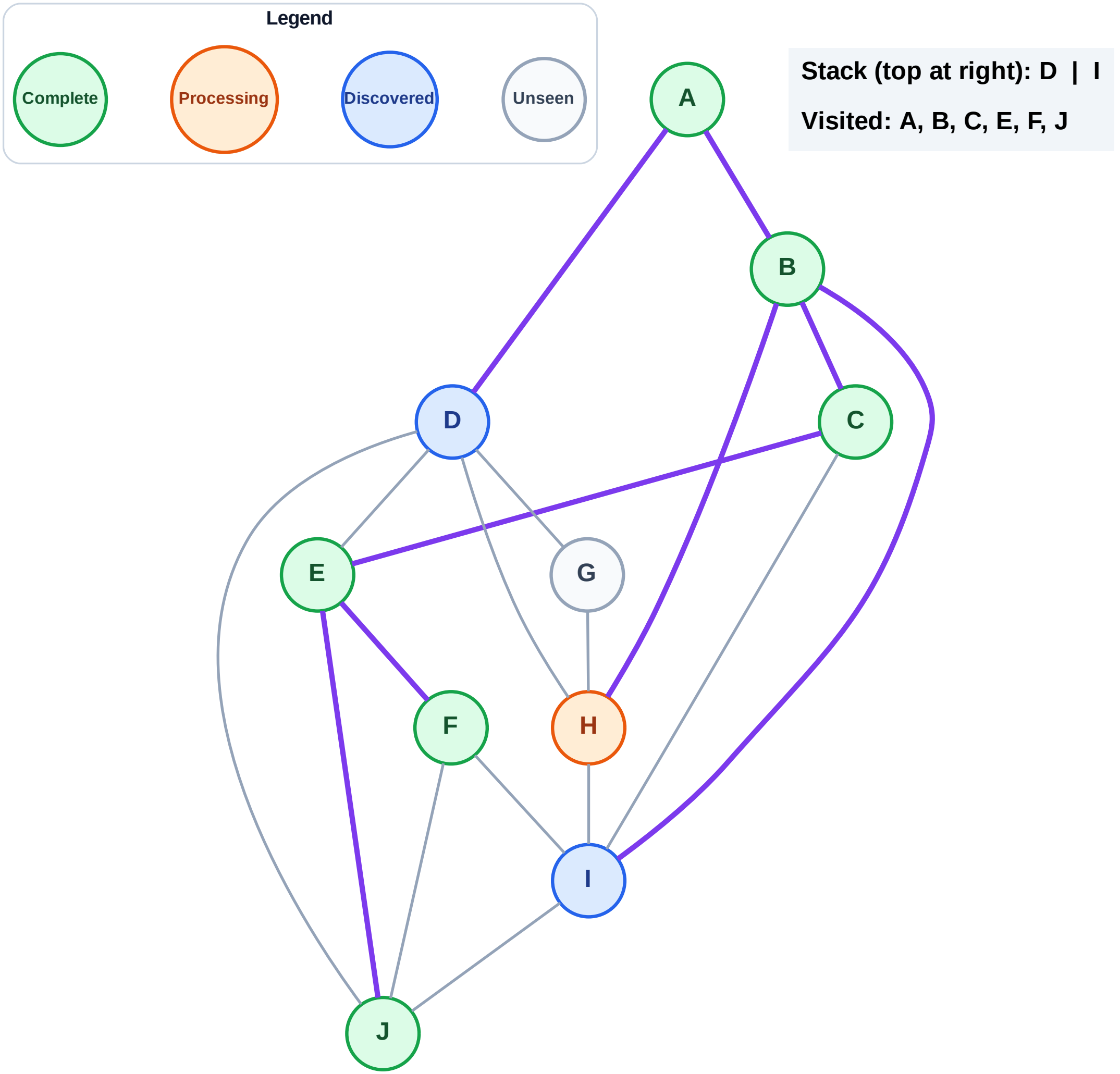


Stack (top at right): D | I | H

Visited: A, B, C, E, F, J

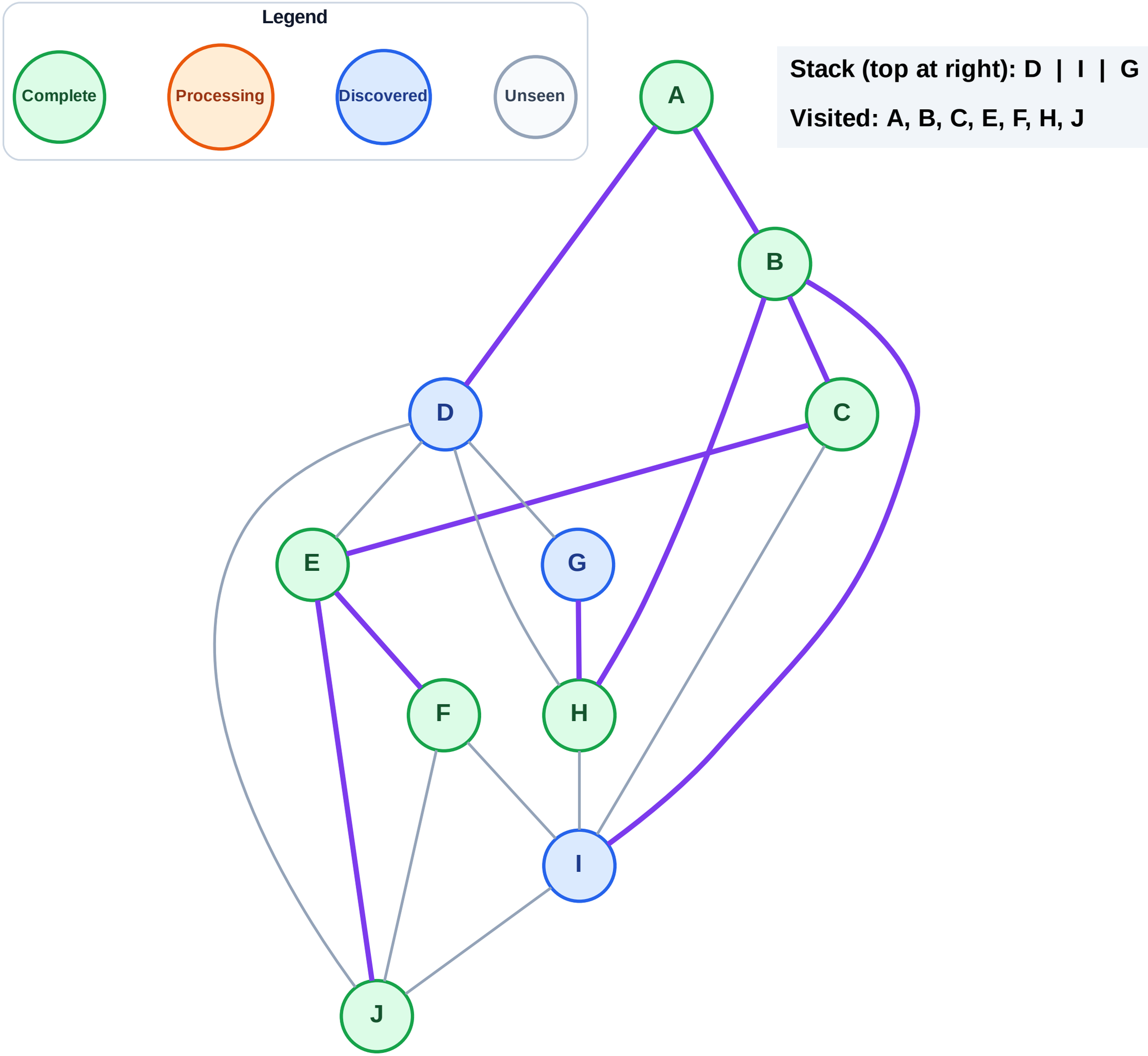
DFS Traversal

Step 14: Process node H



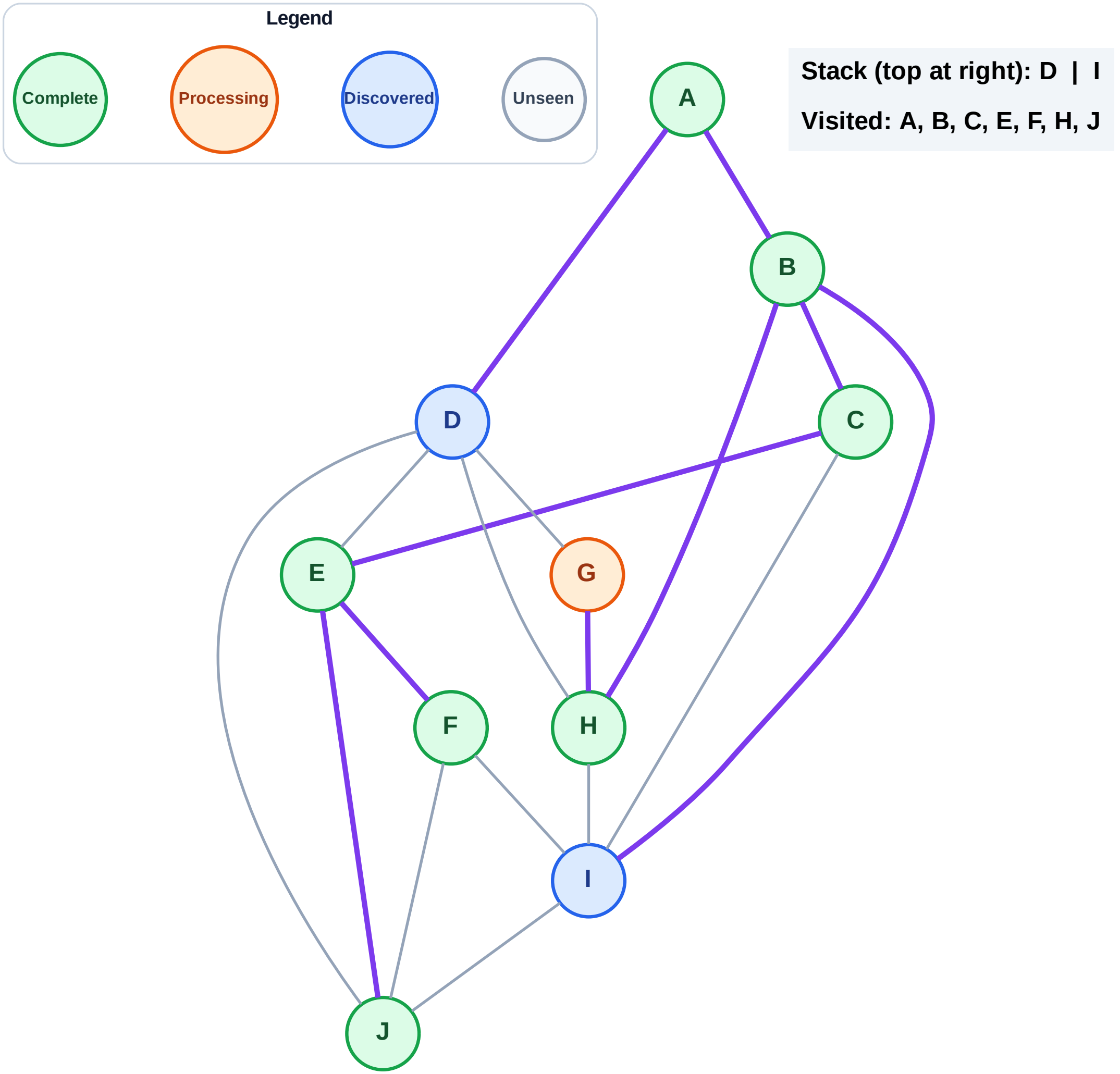
DFS Traversal

Step 15: Discover G from H



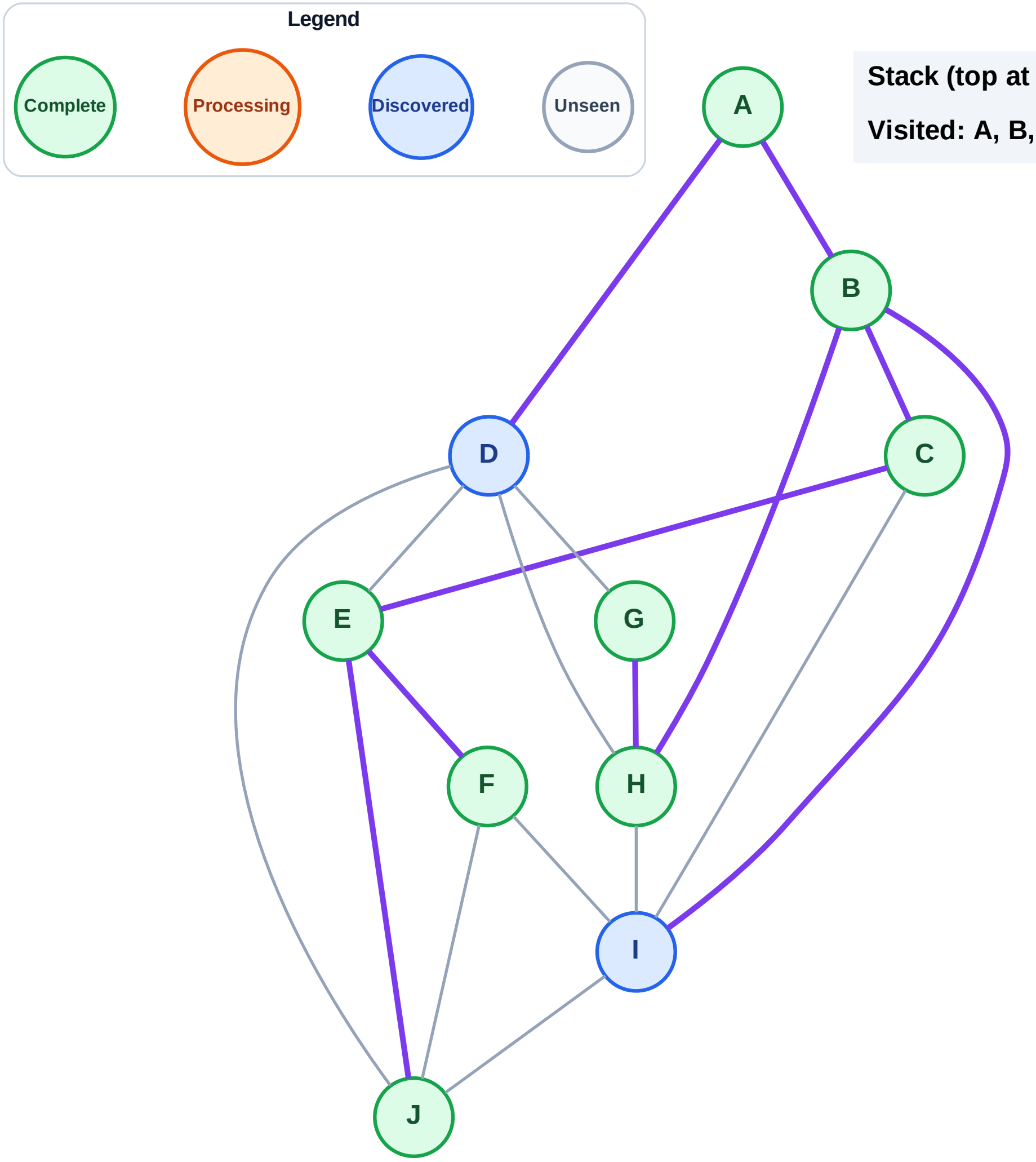
DFS Traversal

Step 16: Process node G



DFS Traversal

Step 17: No new neighbors from G

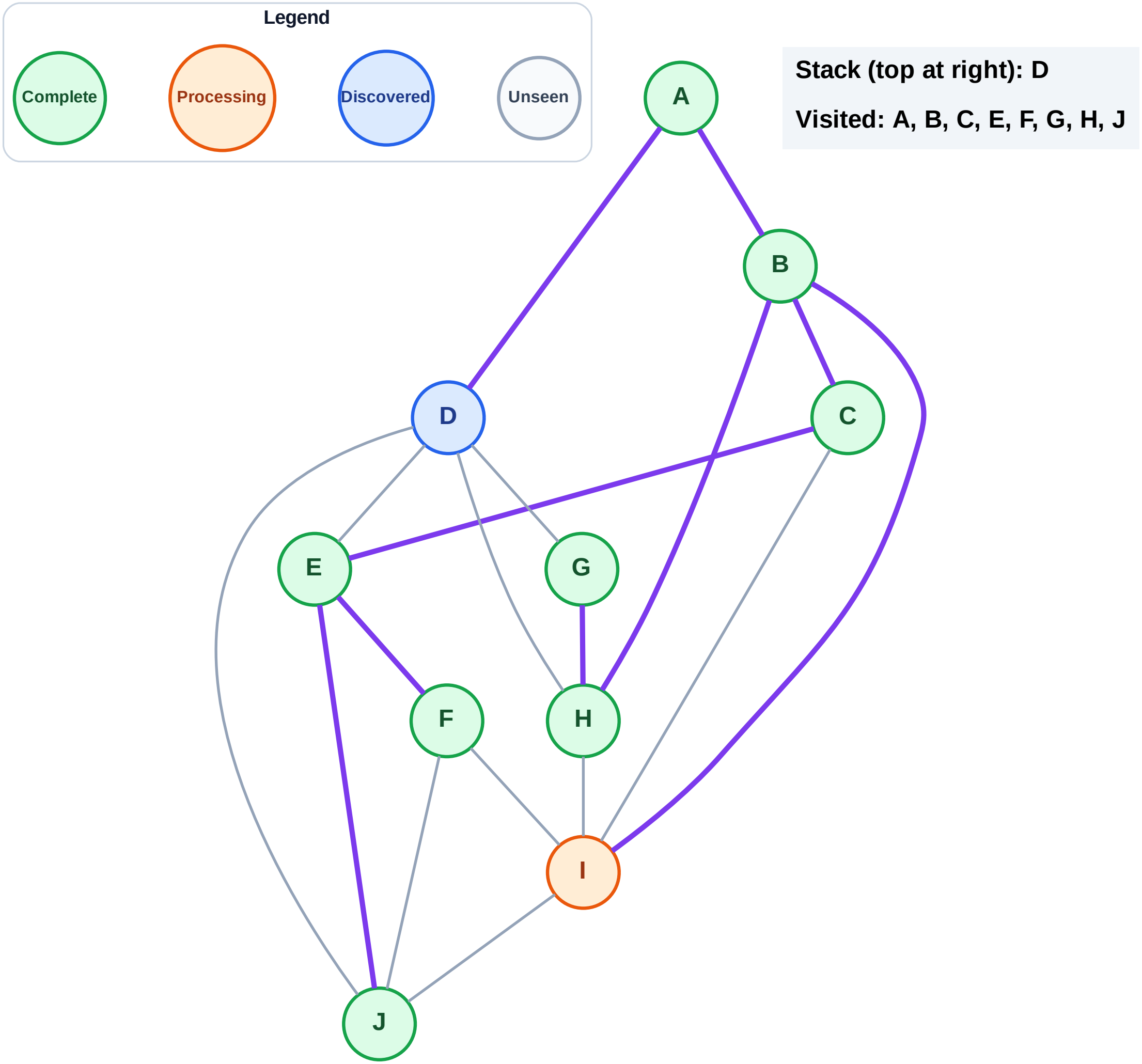


Stack (top at right): D | I

Visited: A, B, C, E, F, G, H, J

DFS Traversal

Step 18: Process node I

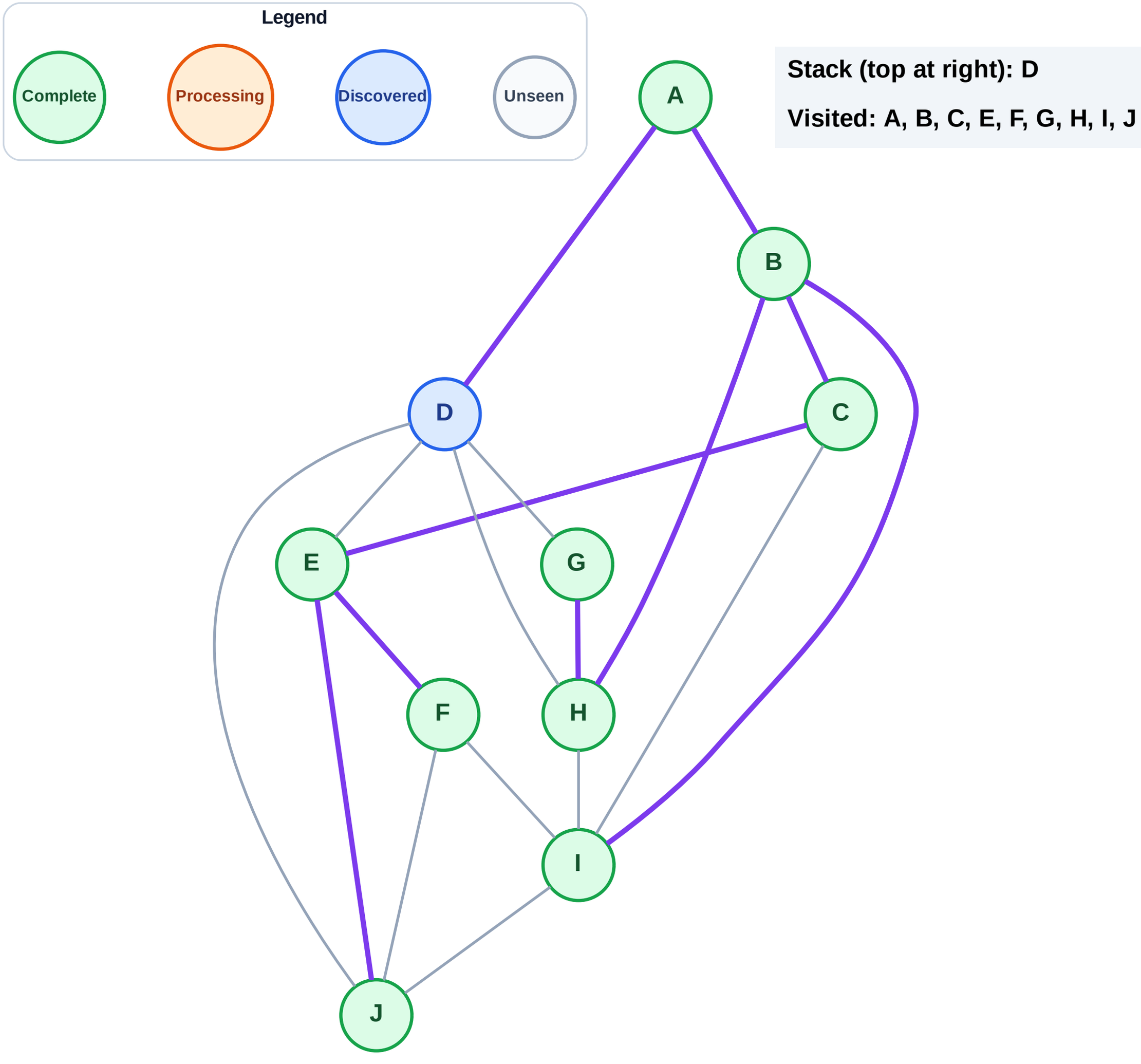


Stack (top at right): D

Visited: A, B, C, E, F, G, H, J

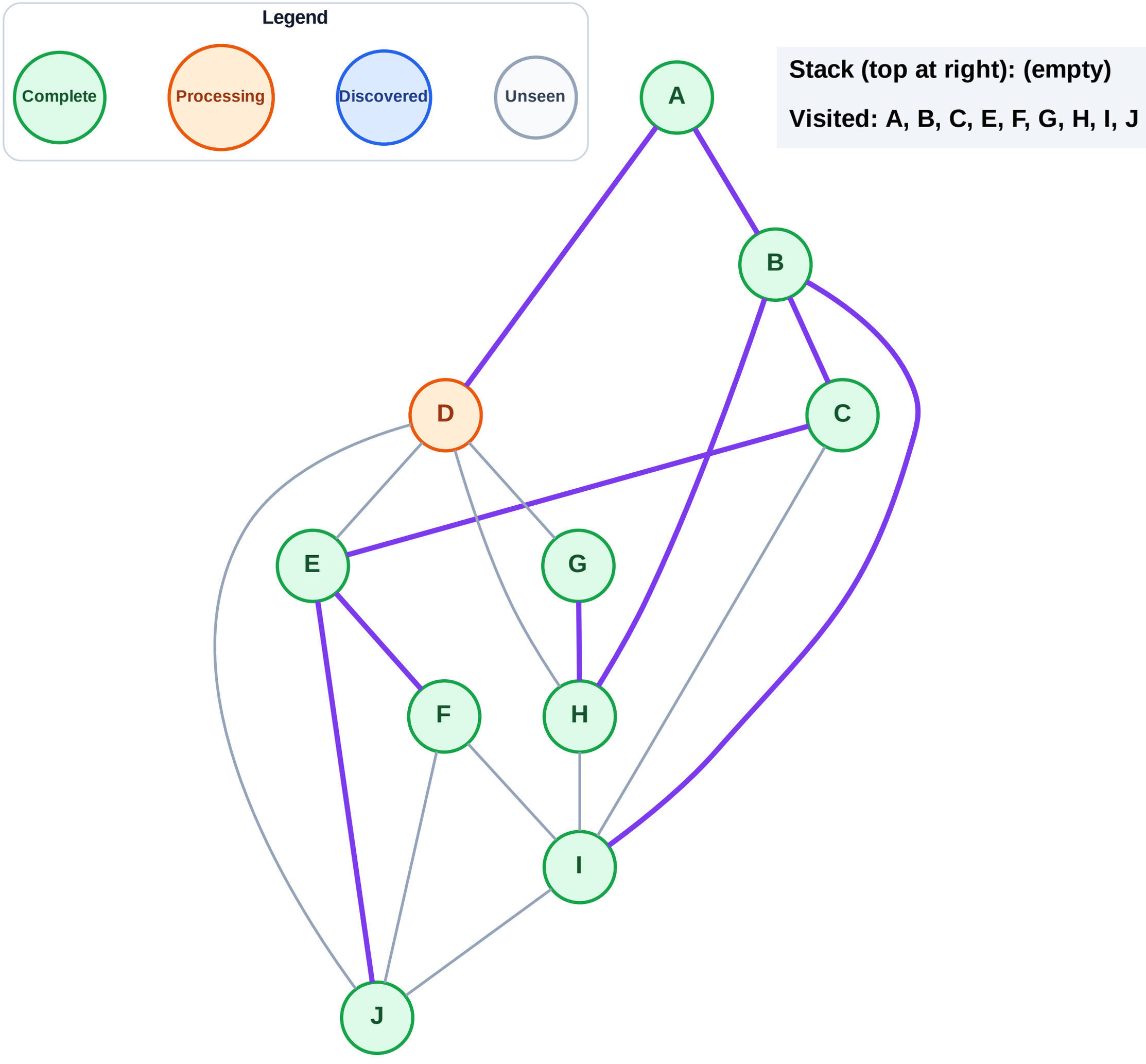
DFS Traversal

Step 19: No new neighbors from I



DFS Traversal

Step 20: Process node D

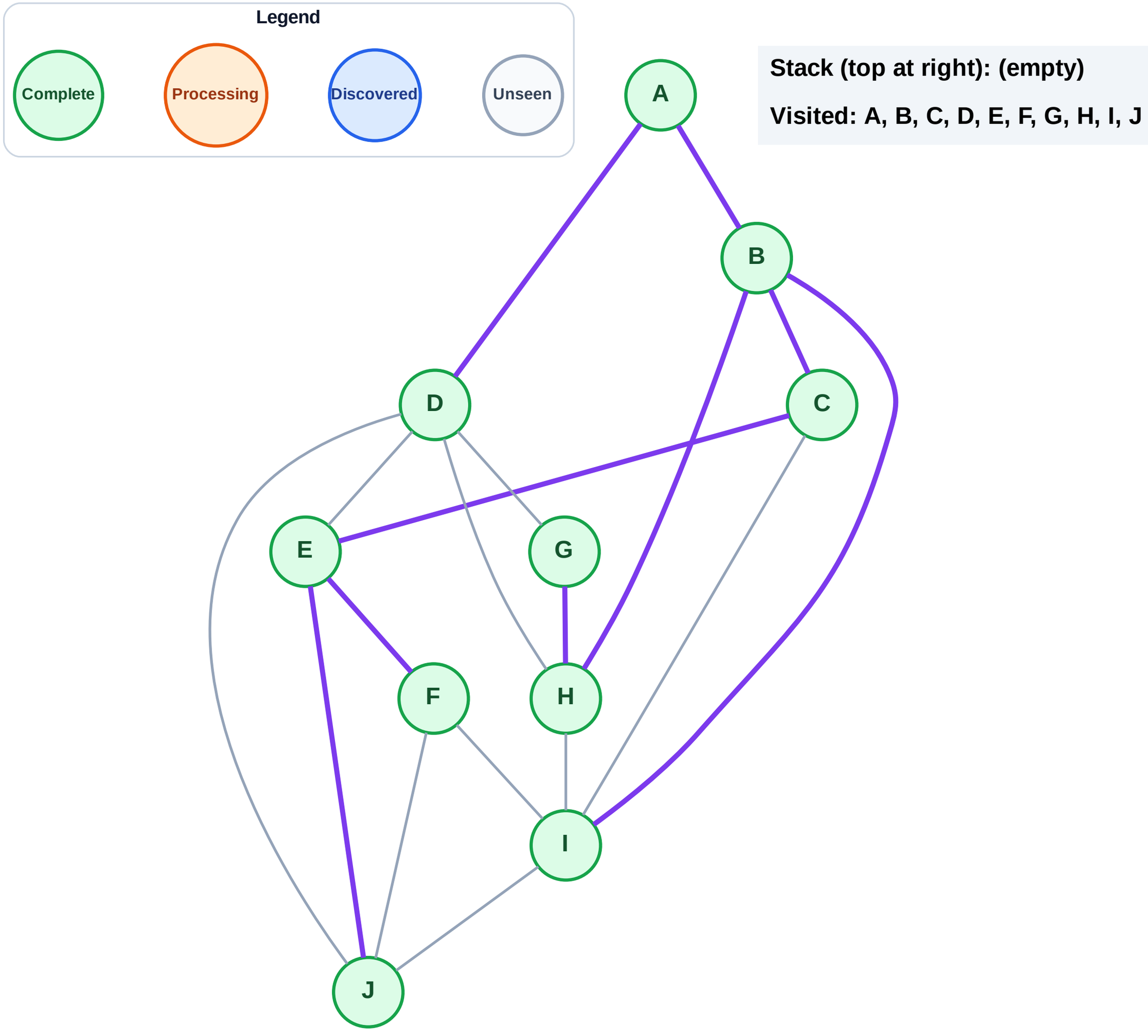


Stack (top at right): (empty)

Visited: A, B, C, E, F, G, H, I, J

DFS Traversal

Step 21: No new neighbors from D



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J